HealthBank

Training Manual

For version 1.0

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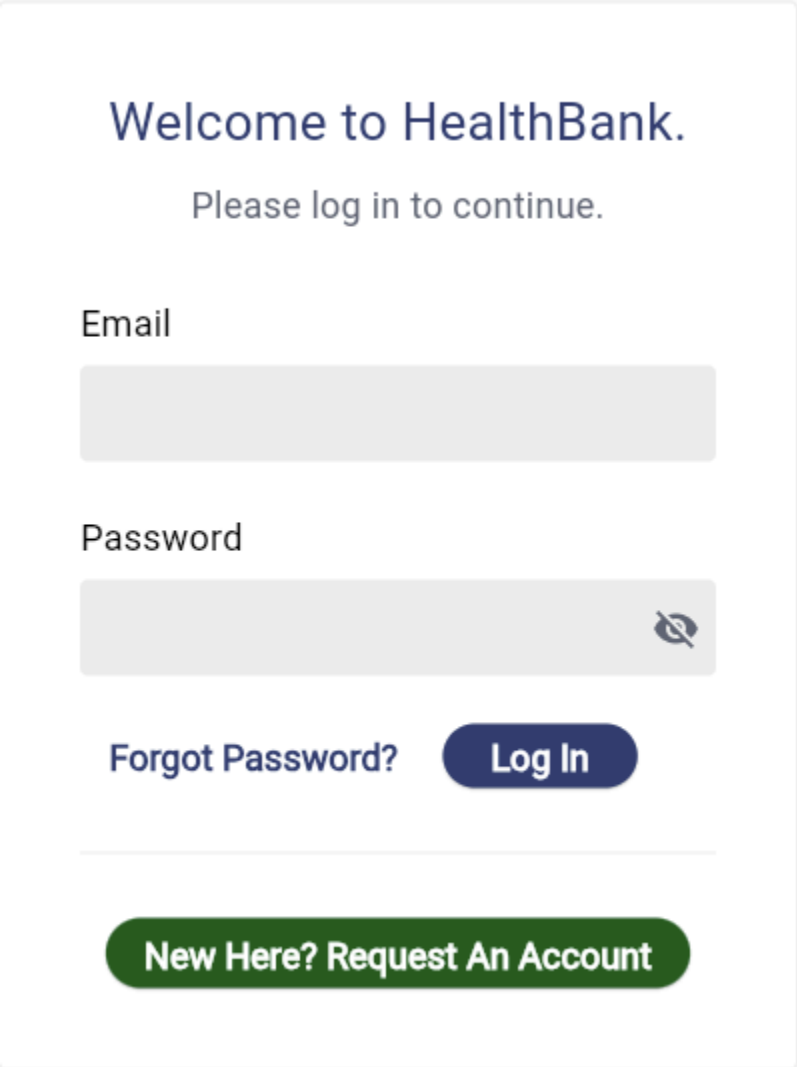
UI Familiarization

Login Page

Upon going to the website, you will be greeted with the login page. If you have an account already registered, you will need to enter your email and password in order to log in to your relevant page.

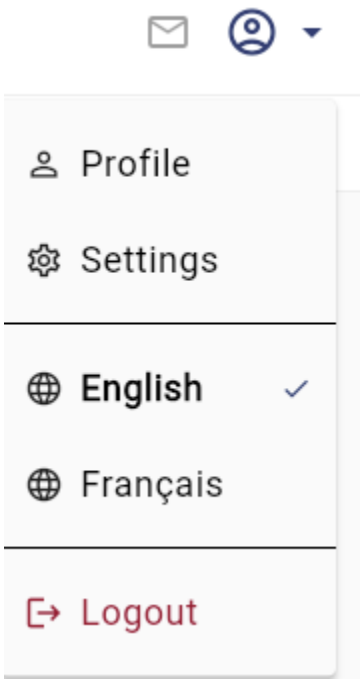


If you do not have an account set up yet, you can click on the “New Here? Request an Account” button found below the password entry section.

This takes you to a section where you enter your first and last name, your email address, and where you select the role you are (Participant, Healthcare Provider, or Researcher). This request is sent up to an administrator to approve or deny your request, which may take some time to complete.

Profile Menu

After logging in, you will be taken to your dashboard. The common UI element between all users is the profile section. On the top right of the page, you will see a ‘mail’ button and a ‘profile’ button; clicking on the latter will show the following menu.



You will now have a few options. The profile button will take you to your profile information page, where you can see things like your name information, your email, your birthday, and gender.

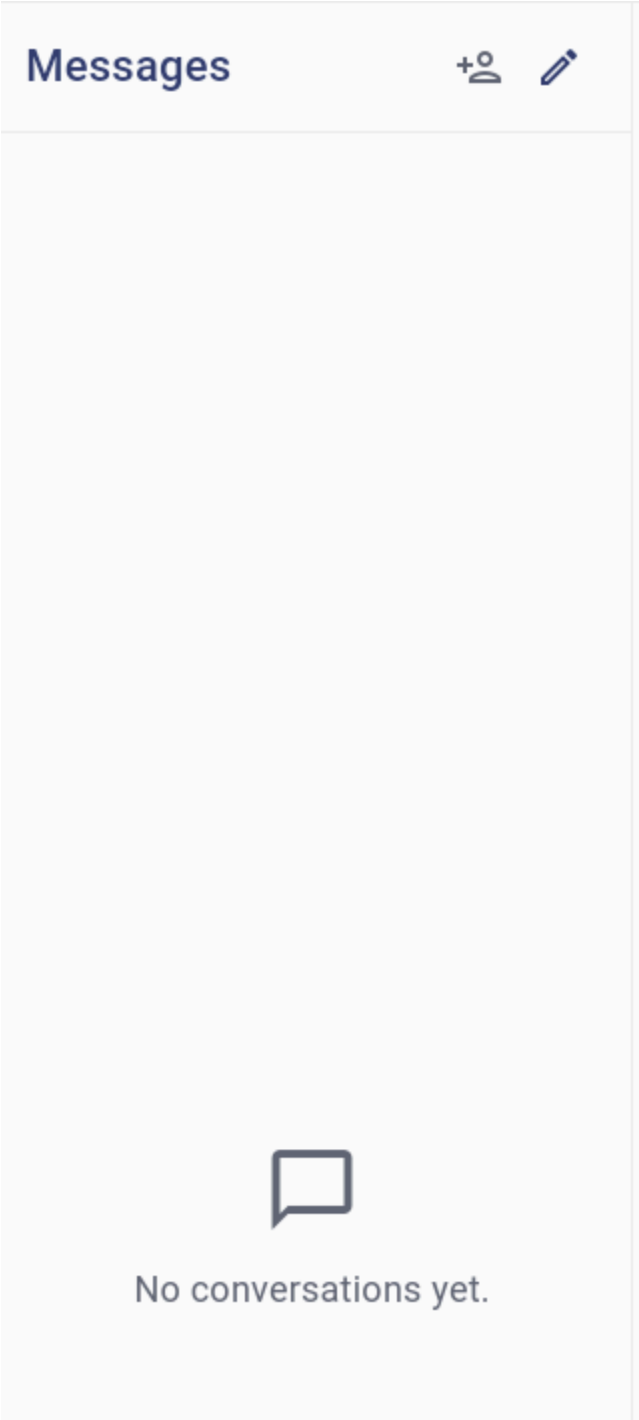
The settings page takes you to the Two-Factor Authentication settings, where you can either set up Two-Factor Authentication with a second device or disable the functionality if wanted; though it isn’t recommended for researchers to do this.

You can also request to have your account deleted, which will be sent to an administrator for approval.

Below you can see the language settings. As of version 1.0 there are localizations of English and French. Selecting English will change the text of the website to English, and French will change it to French; English is set as the default. Questions in surveys will not be changed, and must be written bilingual.

Messaging

The final tab we have is the Messages tab. In here, you will be able to message members of this research project.



The two options of “send friend request” and “new message” are available on the top section. Whereas your message history will be contained below.

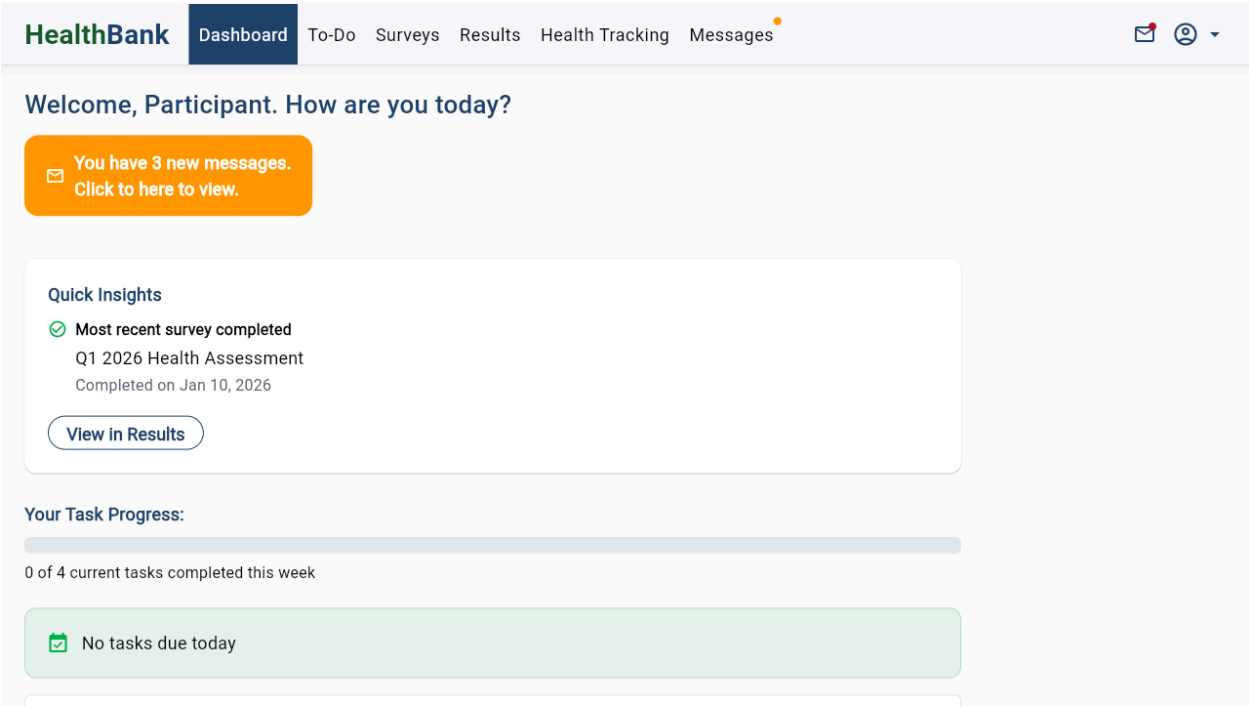
Send Friend Request will prompt you to enter an email address, and will request the requisite member to add you as a ‘friend’ in order to communicate via the messaging system in the future. This option is intended for researchers to be able to reach out to members participating in this research project, but could be used by participants to speak with each other or to Healthcare Providers.

Participant Overview

This section covers the features and pages available to participants. Participants are the individuals enrolled in research studies who provide health data through surveys, health tracking entries, and other activities. Upon logging in, participants are taken directly to their personal dashboard.

Dashboard

The participant dashboard provides a quick summary of account activity. At the top of the page, a Quick Insights card shows an overview of recent health tracking activity, including the date of the last check-in and whether any tasks are outstanding for the current week.

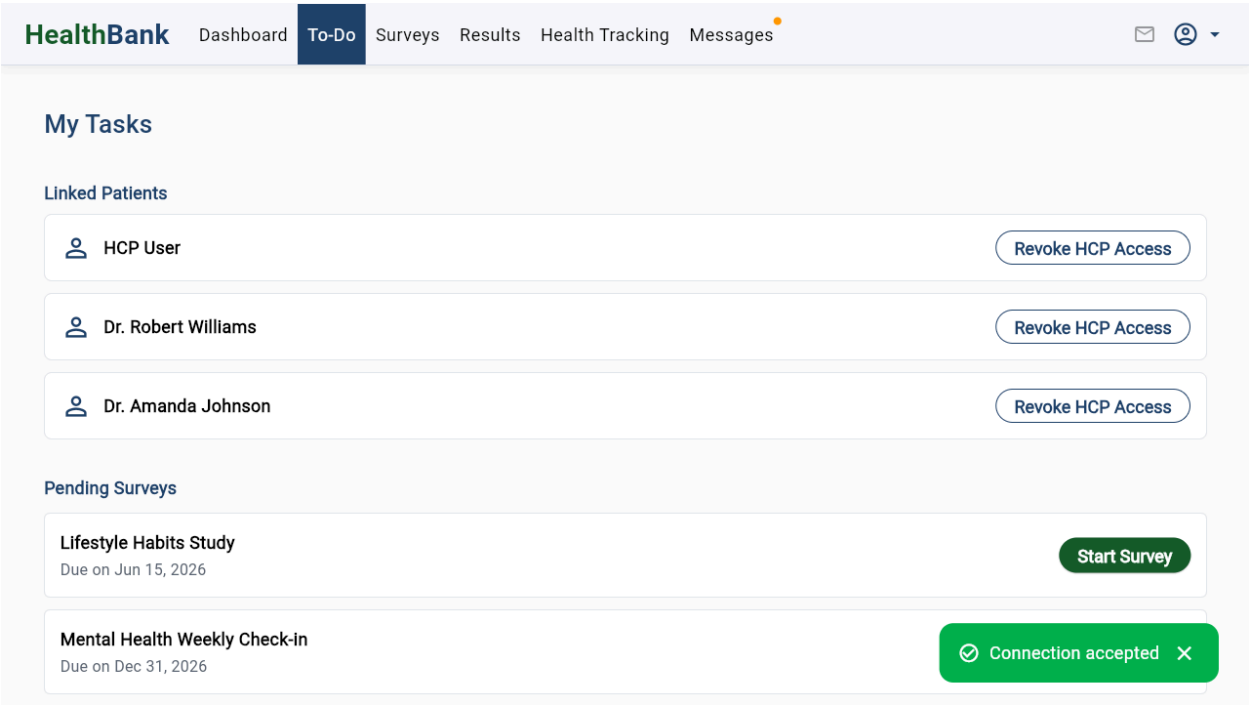


Participant Dashboard

Below the insights card, a task progress section shows how many outstanding survey tasks are due and how many have been completed. Each task is listed with its title, due date, and a button to navigate directly to that survey. If the participant has not yet recorded their baseline health snapshot, a notification banner will appear prompting them to complete it.

Tasks

The Tasks page lists all survey tasks that have been assigned to the participant. Tasks are displayed in a card layout showing the survey title, a short description, and the due date. Tasks are grouped into "Pending" and "Completed" categories.

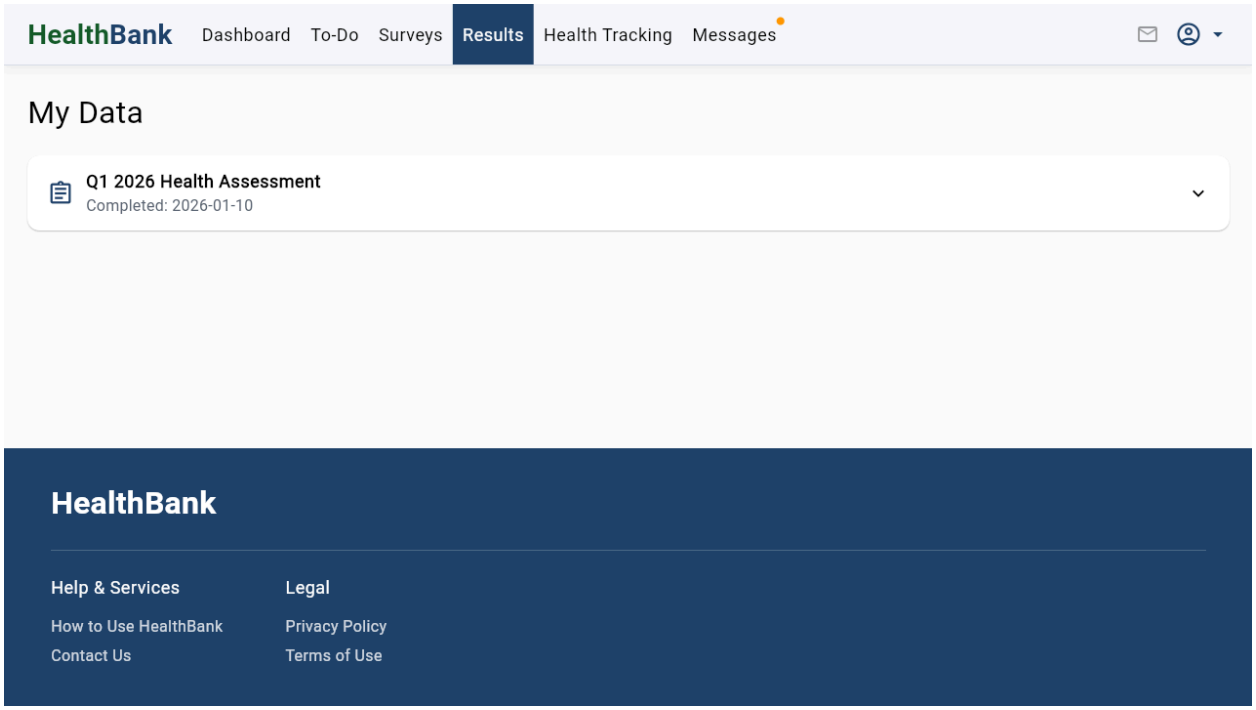


Participant Tasks page

Tapping or clicking a pending task opens the survey form directly. Once all questions are answered and the form is submitted, the task moves to the Completed category and the task count on the dashboard updates automatically.

Results

The Results page allows participants to review the responses they have submitted for completed surveys. Each completed survey is shown as an entry with the survey name and submission date. Selecting an entry opens a read-only view of all questions and the participant's recorded answers.

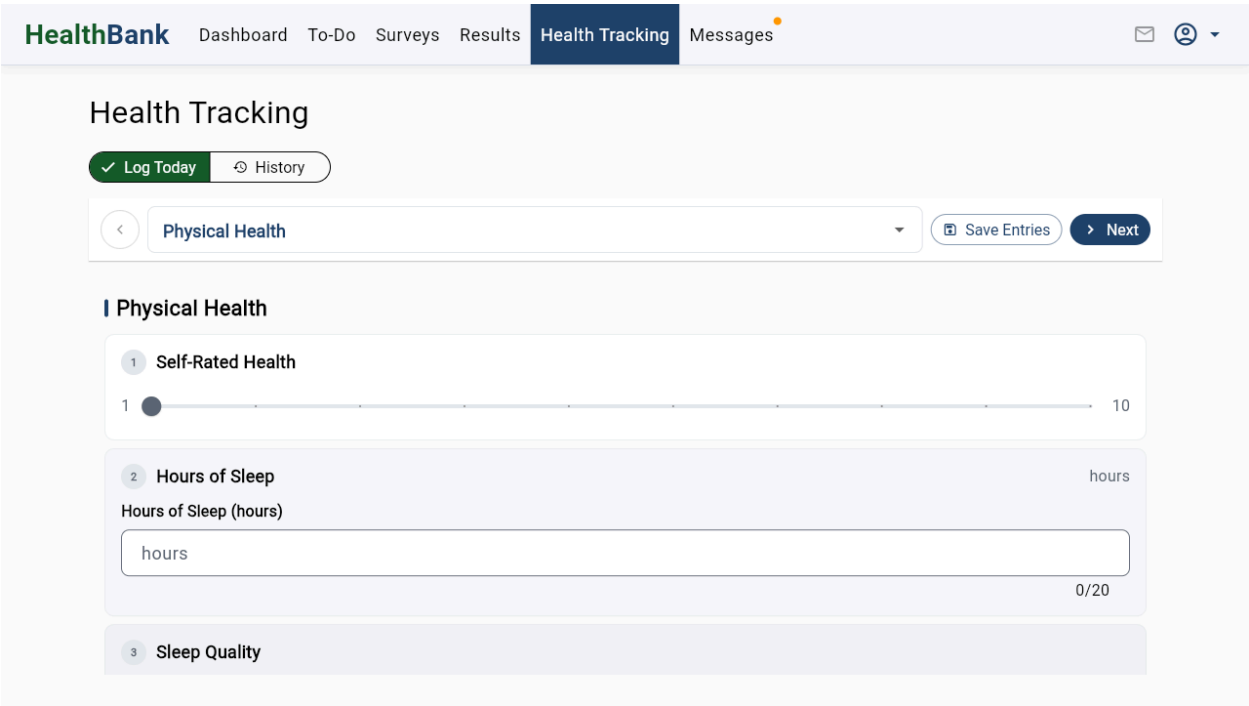


Participant Results page

This page is view-only; responses cannot be edited after submission. Participants can use this page to review their own data at any time.

Health Tracking

The Health Tracking page is where participants log their daily, weekly, and monthly health metrics. The page is divided into two views, selectable via a segmented button at the top: "Log Today" and "History".

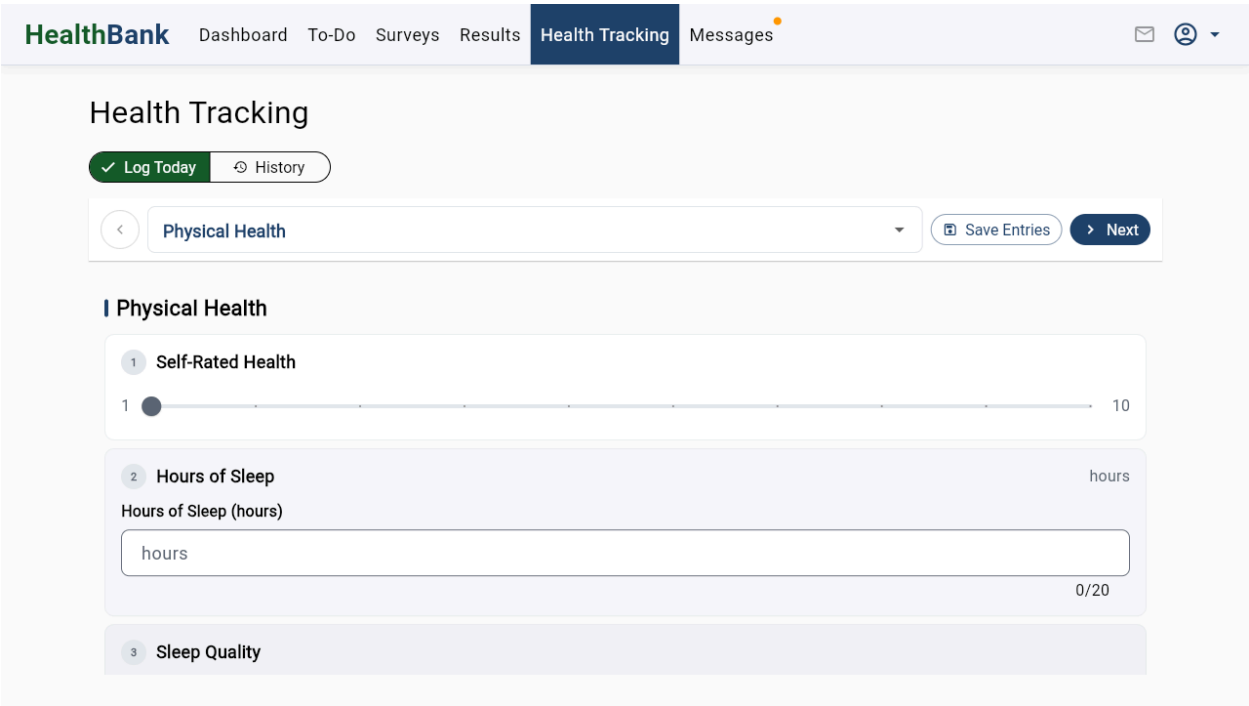


Health Tracking — Log Today view with category selector and metric cards

The Log Today view presents all health metrics that are due on the current day. Metrics are organised by category — such as Physical Health, Mental Health, Nutrition, and Housing Stability — using a dropdown selector at the top. Back and Next buttons allow navigating between categories.

Each metric is presented as an entry card. The appearance of each card depends on the metric type configured by the administrator:

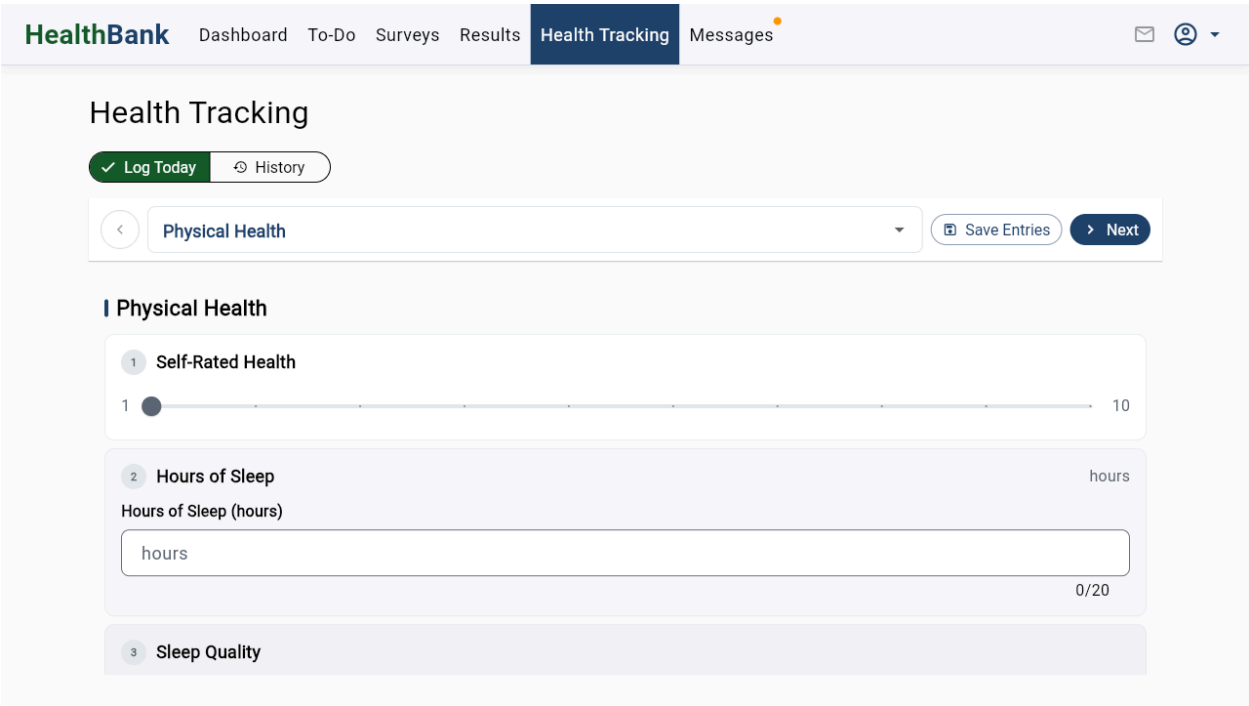
- Scale metrics (e.g. "Self-Rated Health: 1–10") display a slider control.
- Number metrics (e.g. "Hours of Sleep") display a numeric input field.
- Yes/No metrics (e.g. "Medications Taken as Prescribed") display a Yes / No toggle.
- Single-choice metrics display a dropdown list of options.
- Text metrics display a free-text entry field.



Different metric input types — slider, number field, and yes/no toggle

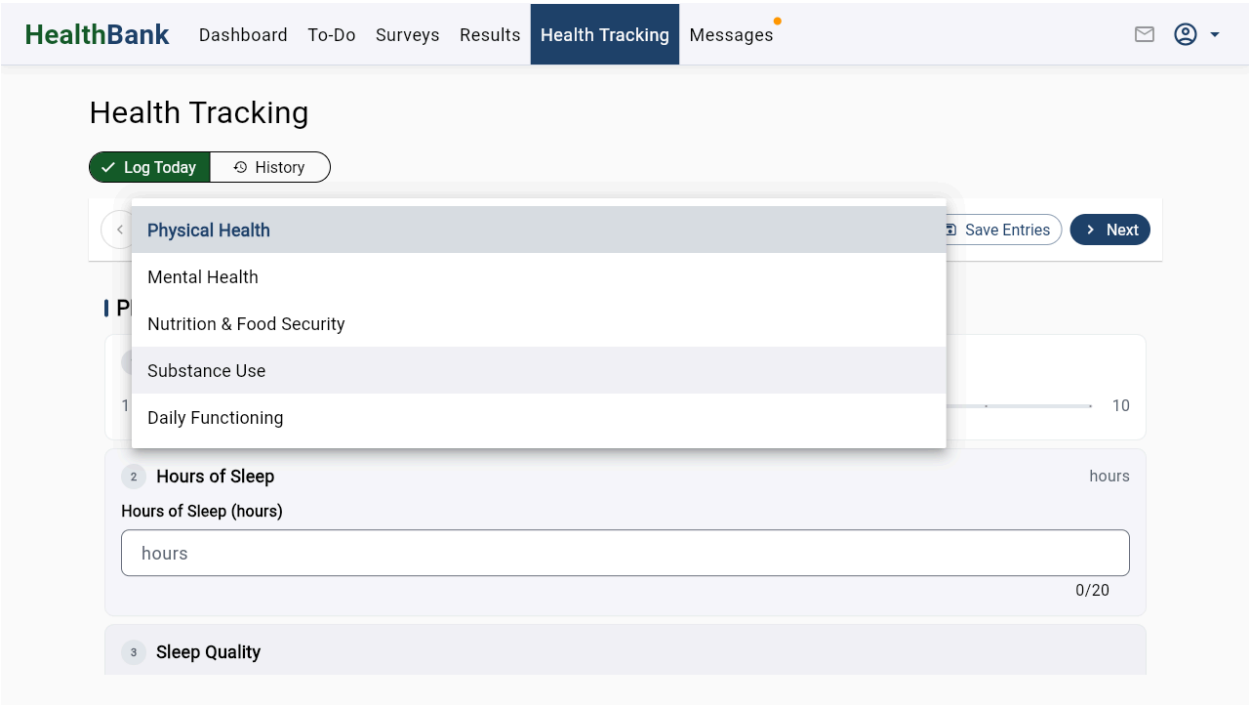
Daily metrics appear every day. Weekly metrics appear on Sundays, and monthly metrics appear on the first day of each month. A "Save Entries" button at the top of the form submits all filled-in entries at once. A success or error notification confirms whether the save was successful. Previously saved values for the current day are pre-filled when the page loads.

If no baseline snapshot has been recorded, the Log Today view shows a prompt at the top of the page to save the current entries as a baseline. The baseline represents the participant's health state before receiving stable housing and is used as a comparison reference in research analytics.



Baseline snapshot prompt shown when no baseline has been recorded

The History view allows participants to review their past entries for any metric. A line chart displays the participant's recorded values over time. The date range can be adjusted using the date picker controls above the chart.



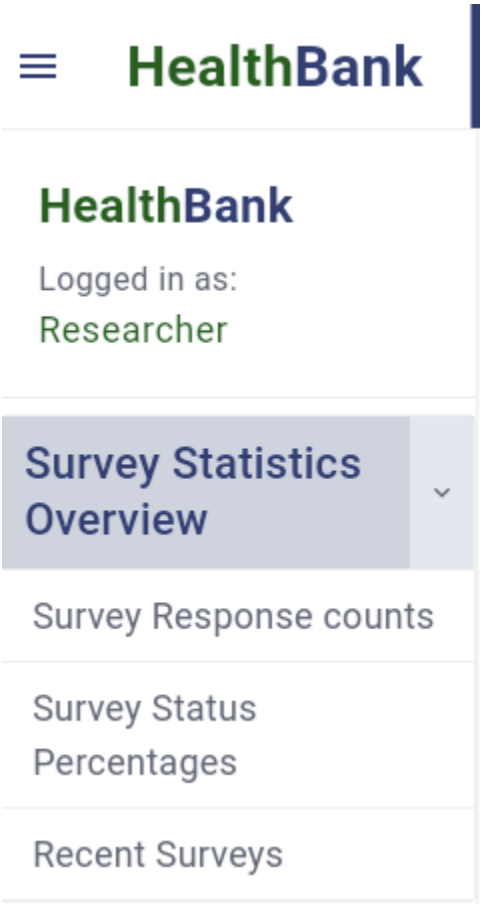
Health Tracking History view with line chart

Researcher Overview

Dashboard

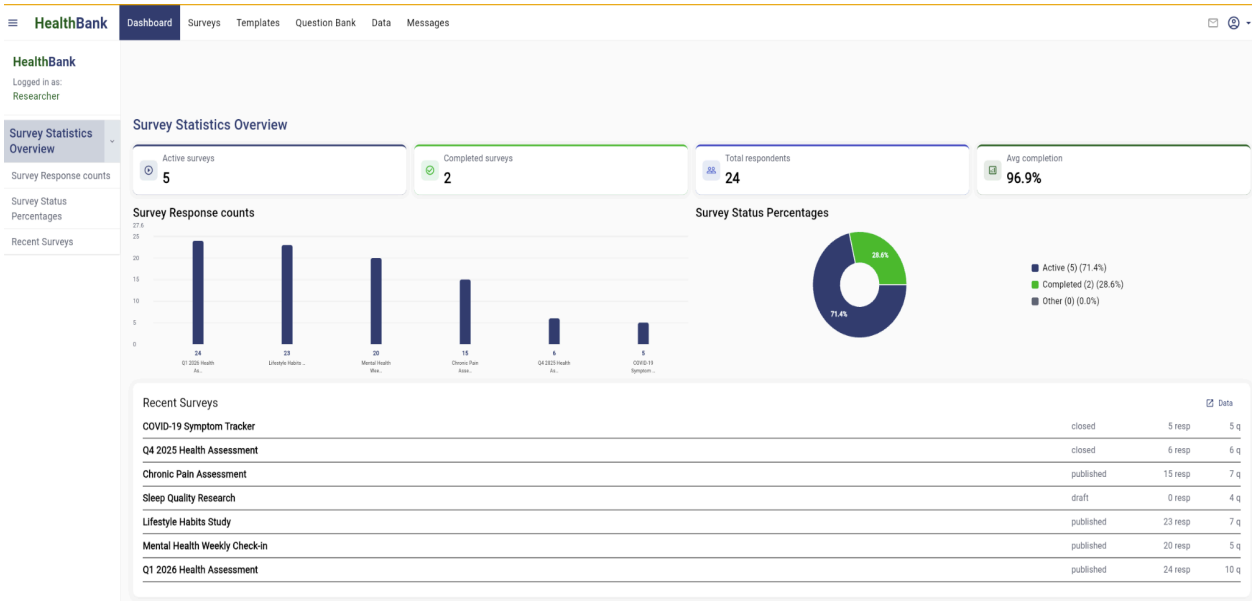
Upon logging in the Researcher will be taken to a dashboard. You may see a large amount of information on the dashboard, but the important buttons are on top of the board. These are the ‘power’ buttons:

These buttons will take you to different sections of functionality for the Researcher role. The default on logging in is the “Dashboard” option. On the left side you will see the following information:



This is your viewing section for the dashboard. Showing you what information is displayed in front of you, and clicking on it will put you in the requisite area for said information.

As a whole, the Dashboard looks like the following:



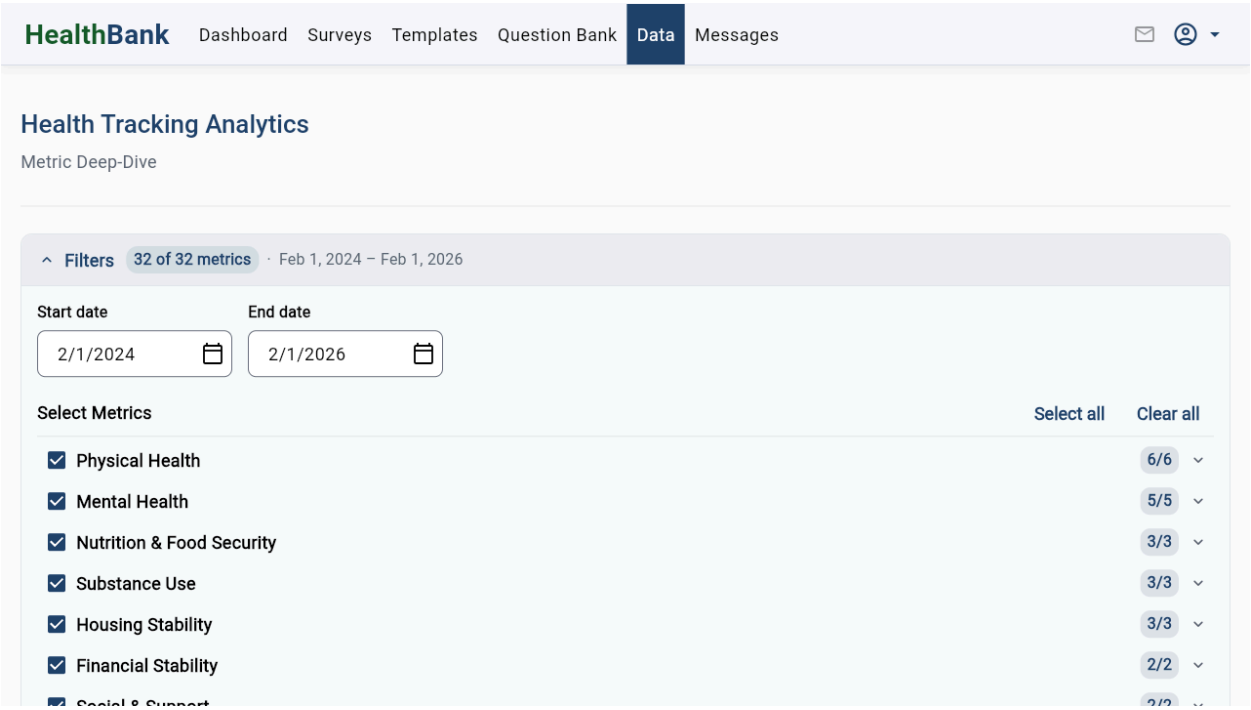
All information in the above dashboard is seeded, and does not reflect any answers to surveys by participants.

Health Tracking Tab

This section describes the Health Tracking analytics feature added to the Researcher area. It supplements the existing Researcher Overview section and should be inserted after the "Data" subsection.

Health Tracking

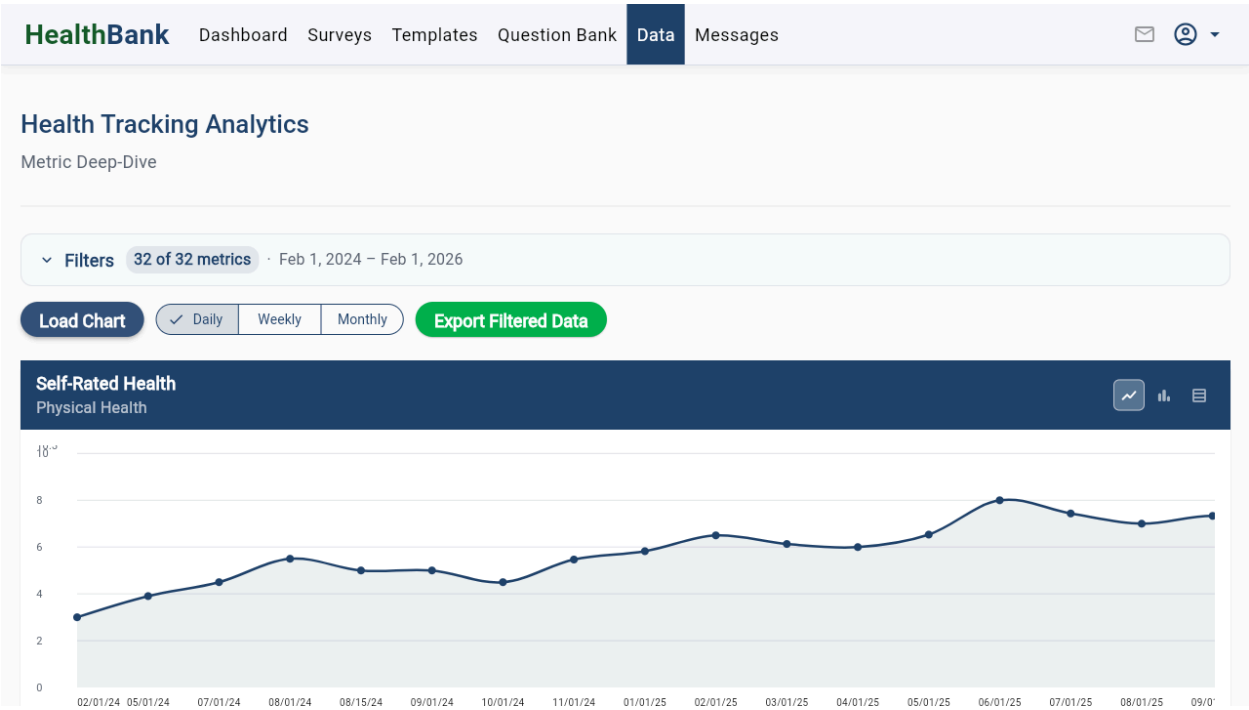
The Health Tracking page provides population-level analytics across all participant health tracking data. Access it by selecting "Health Tracking" from the researcher navigation menu (found under the Data tab).



Researcher Health Tracking page — filter panel showing 32 of 32 metrics selected

At the top of the page is a collapsible filter panel that controls which data is charted. It shows a summary of the currently selected metrics and date range. Expanding it reveals the full metric selection and date range controls. A granularity toggle switches chart aggregation between Daily, Weekly, and Monthly averages.

Clicking "Load Chart" fetches the aggregated data from the server and displays one chart card per selected metric.



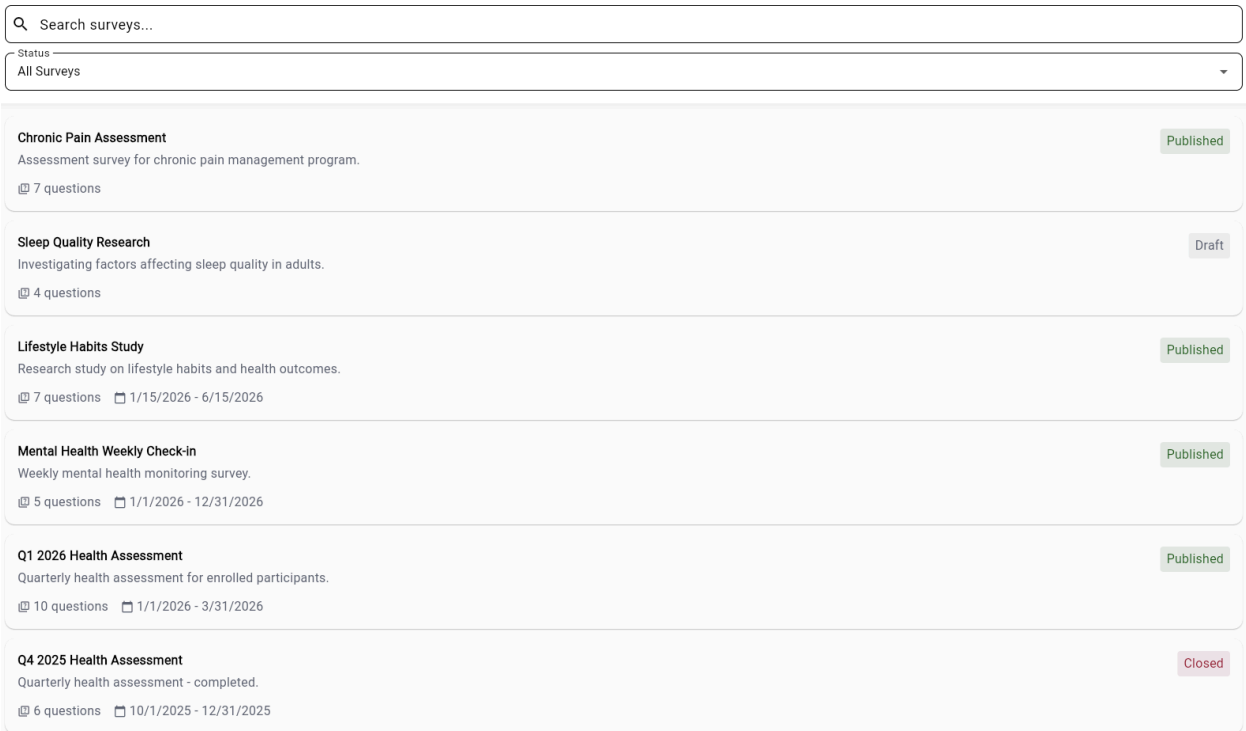
Researcher Health Tracking — chart loaded showing Self-Rated Health over time

Each chart card shows the metric name and category. All data displayed is k-anonymized: individual participant values are aggregated and only group averages are shown. Data points representing fewer than the k-anonymity threshold number of participants (configured in Admin Settings) are suppressed and not plotted.

An "Export Filtered Data" button downloads the currently loaded data as a comma-separated CSV file. The export respects the active metric selection and date range, and all exported data is k-anonymized identically to the on-screen charts.

Surveys Tab

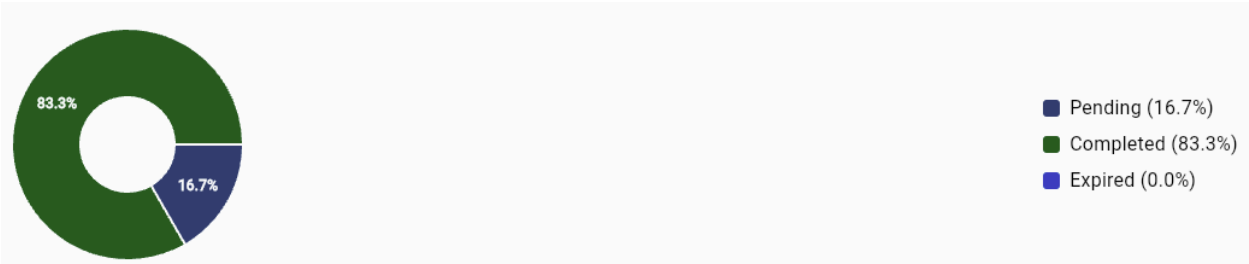
Upon clicking the button labelled ‘Surveys’ at the top of your screen, you will be taken to the page that shows all current surveys.



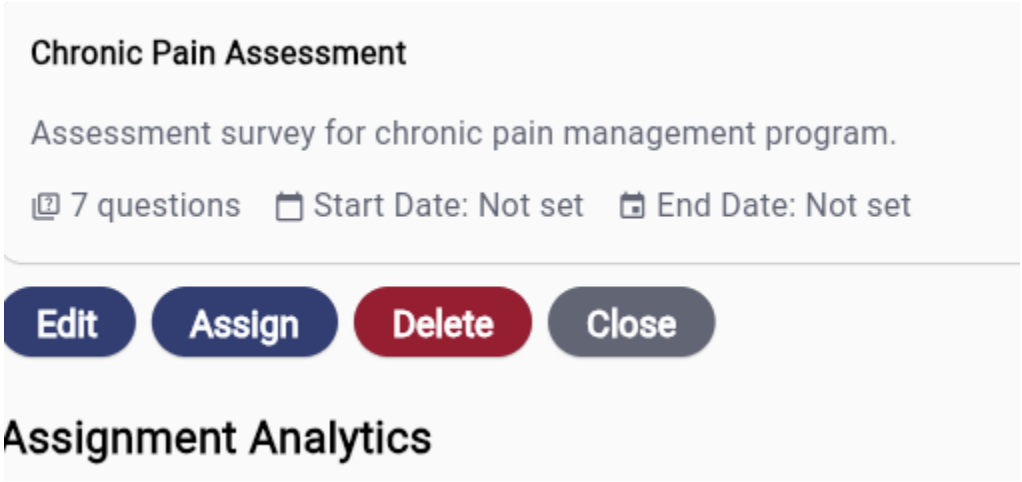
You will have a search function, where you can search surveys by name. A filter, which can currently filter by drafts, published, or closed surveys. Where drafts are surveys not pushed to the participants yet, published are currently active surveys that are gathering information, and closed are surveys that are no longer being answered.

Survey Menu

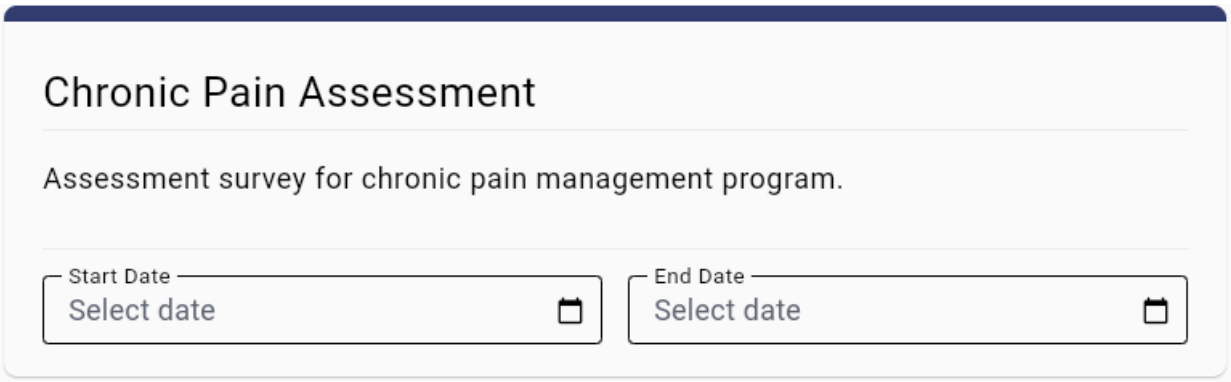
Clicking on any survey will bring you to the Survey Status screen. Where you will gain some information about the survey. Such as the name, question number, start date, and end date. Included is the participant information of the assigned participants, how many have completed the survey, how many have not completed it, and how many have failed to complete it after the survey becomes closed. There will be a small pie chart to show you a graphic of this information:



Above the analytics, there will be four options:

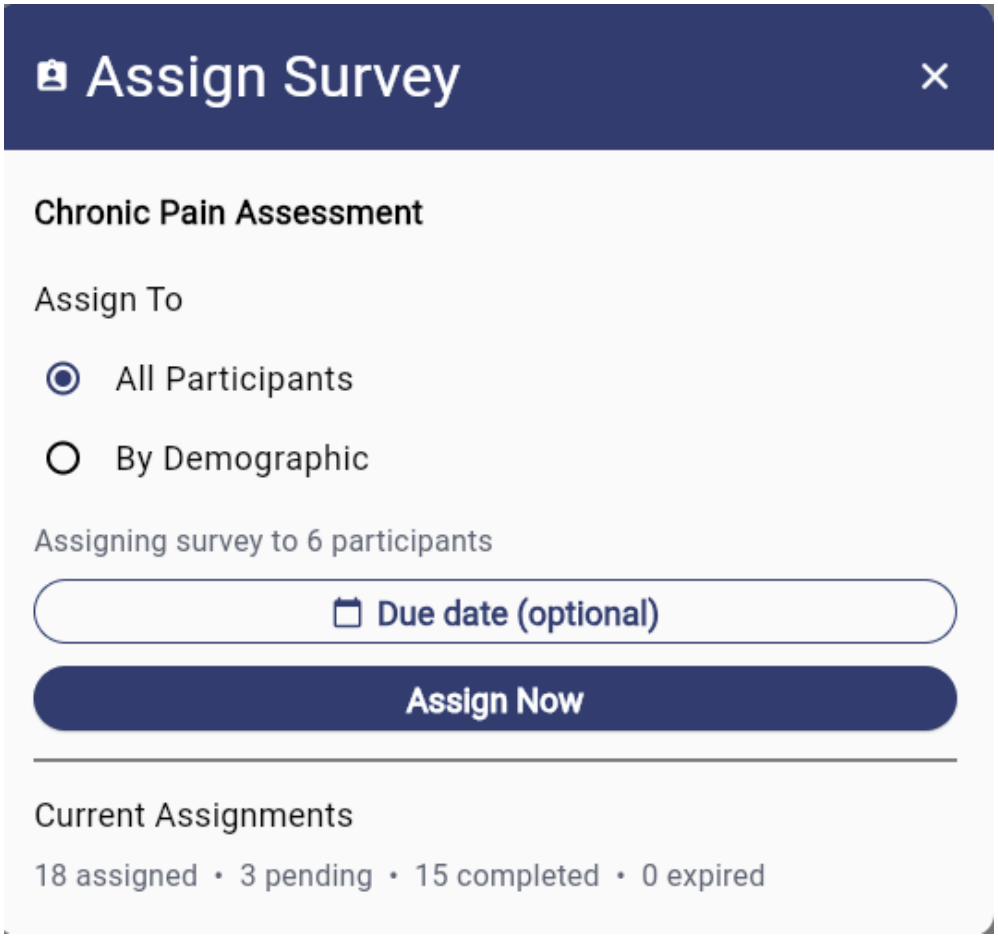


‘Edit’ will take you to a page to modify the questions of the survey, and dates that it is assigned.



The start and end date, when the survey is open, can be found at the top. Whereas the questions of the survey can be modified individually to change the question type (what response the participant gives to each question), the question itself, whether the question is required to be answered on the survey, and to remove the question with the small trash icon on the bottom right of the question menu. Questions can also have their order changed by dragging them in the preferred order.

The ‘assign’ button will give the following popup:



Which can send out the survey to all participants, whether they have completed it yet or not. Or can be narrowed down by the ‘By Demographic’ option, where the following sub-menu will appear:

The current options available are by gender and age, but more could be added in the future. The survey can also be assigned a ‘due date’, where the survey will close on that date.

The ‘Delete’ and ‘Close’ buttons are rather simple.

The delete button will delete the survey permanently, whereas the close button will prevent any further answers for this survey coming in, resulting in the survey being labelled as ‘closed’.

Templates

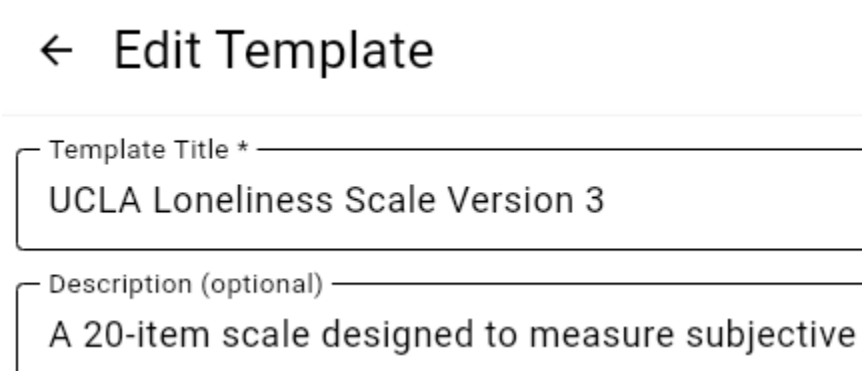
The next option on the top bar is the Templates tab. Slightly different from Surveys, these are usually larger versions of surveys that all researchers will be able to see and use as a baseline for surveys.

The layout is the same as the Survey tab, with a search and filter feature at the top and all templates laid out below. The filters this time are ‘public’ and ‘private’, which will show which templates you have shared with other researchers and which one is linked to your account directly; meaning others cannot see.

Clicking on one of these templates will bring you to the ‘Edit Template’ page.

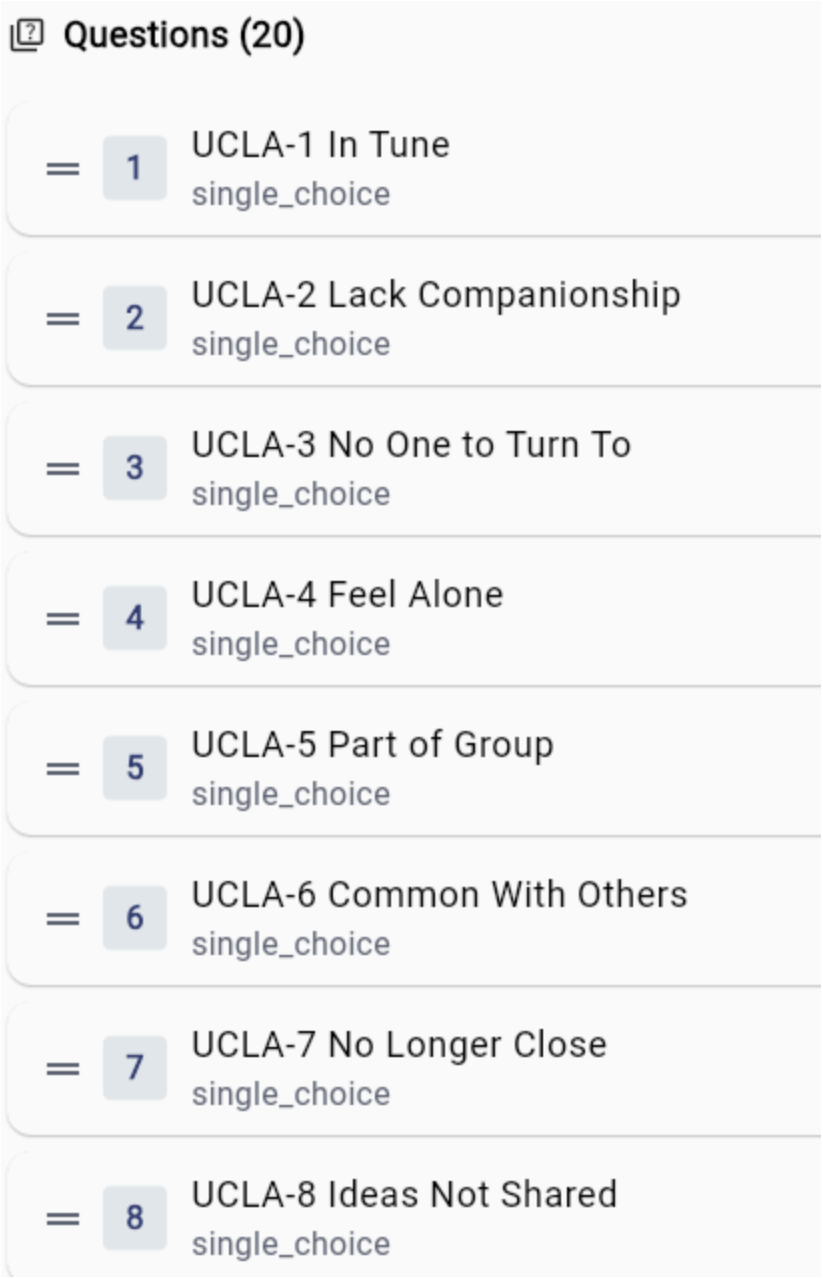
Edit Template

The ‘Edit Template’ page has two sections at the top:



The title is required for the template to be used, whereas the description is optional but recommended.

Questions can be added or taken away from the template, and the questions within the template are listed below in order. The order can be modified by dragging these questions in the preferred configuration.



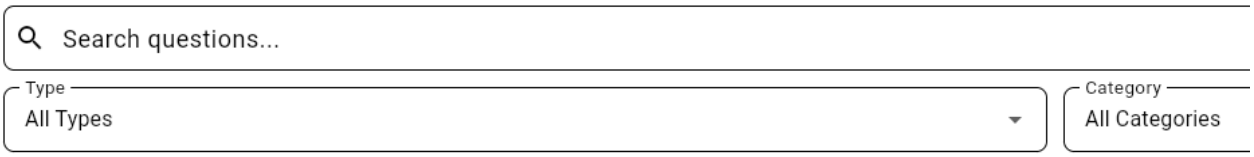
Questions are listed by question number (in said order), the question title, and the option the participant is given.

In this case, all questions are single choice, meaning the participant will pick one answer from a pre-existing set.

On the right side of the page, a green button labelled ‘Add Questions’ will be visible. Clicking on this option brings you to the ‘Select Question’ page.

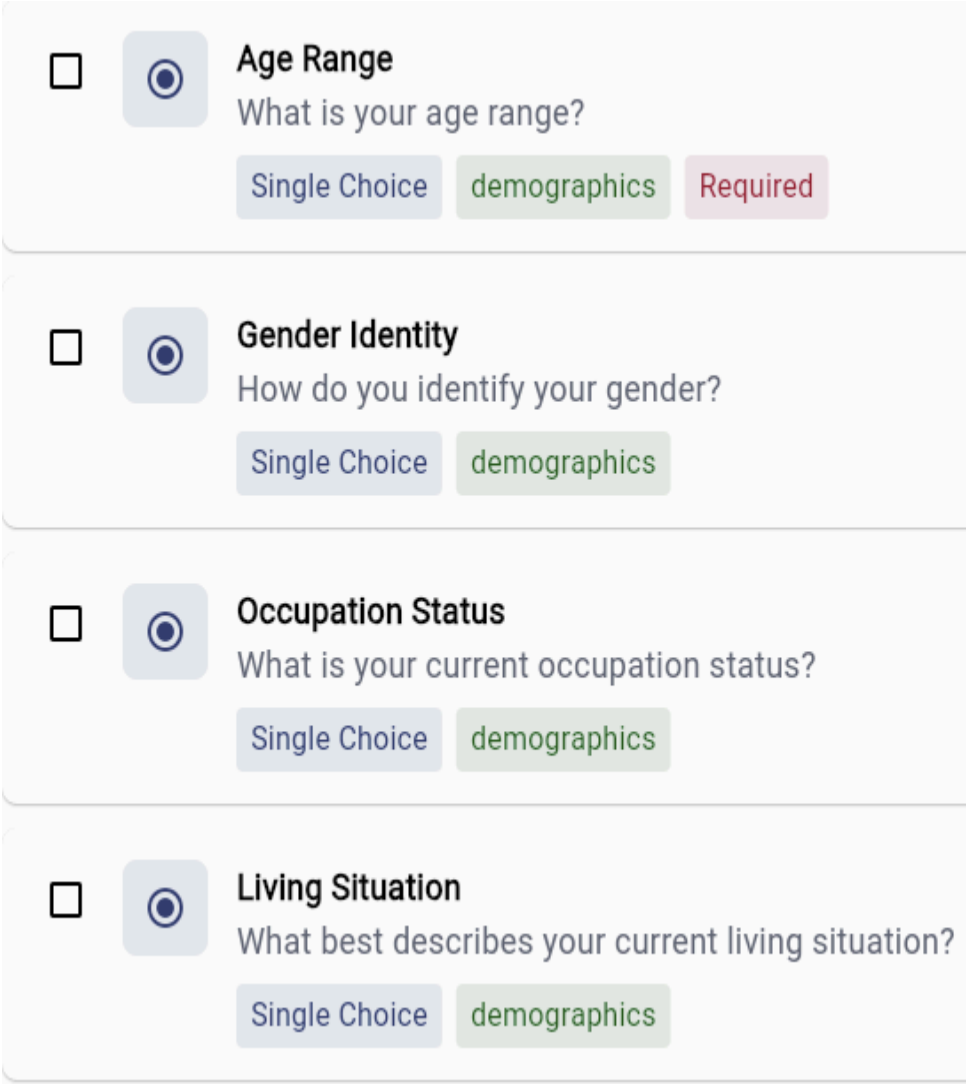
Select Questions

The select questions page has three different options at the top:



You can search questions, filter the questions by types of answer, and filter by preexisting categories such as demographics, lifestyle, or mental health.

Below will be a long list of questions:



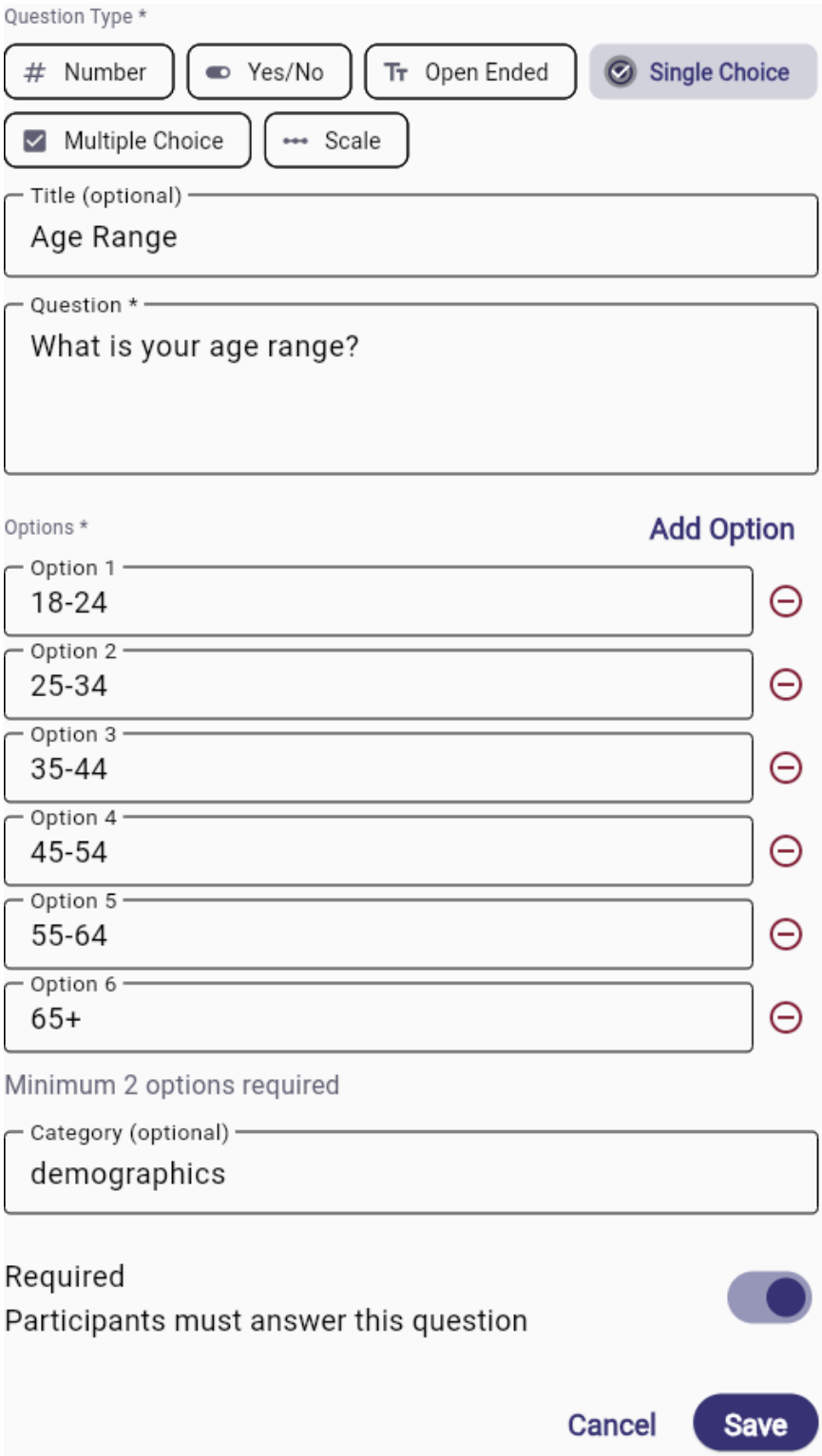
Each question has information on it. The title, the question itself, answer type, category, and requirement.

You can select any number of questions, then click the ‘Add’ button found at the bottom of the page.

The number next to it is the amount of total questions on the template. The questions are taken from the question bank, and are not able to be modified within the template options. After finishing your modifications, you will return to the ‘Edit Template’ tab. Where you can hit the ‘Save’ button on the top right of the page.

Question Bank

The fourth section of the top bar is the question bank. This is where all pre-existing and created questions can be found and modified. The same three options for searching are found at the top, a search bar, as well as filtering by type and category. If you click on a question, in this example we'll use Age Range, you will get the following popup:



This is the edit question menu. Where you can choose from a multitude of selections. The question type, found at the top, dictates how the participant will answer said question.

You also have the title and question, which is what the participant will be prompted with for answering.

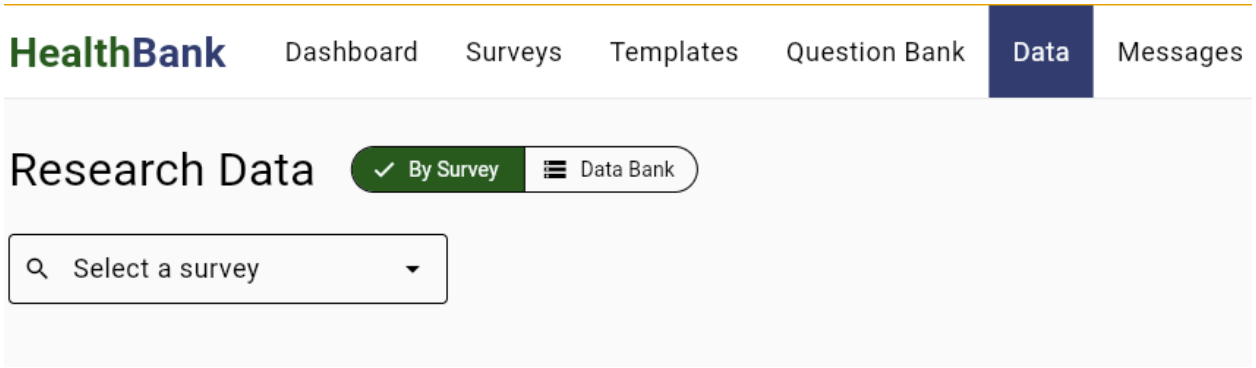
The options, in this case single choice, will be dictated by the researcher. This question has age ranges for the participant to answer with. Below is the category, which can assist with finding this question in the future. And

requirement option, whether the question can be optional for the participant to answer. After all modifications are finished, hit the ‘save’ button and the question will be successfully modified.

Returning to the Question Bank page, at the bottom right of the screen is a button called ‘New Question’. This will take you to a similar popup as the one above, just without any information added in. Once information is added, and all options selected, the question will be added to the question bank for future use.

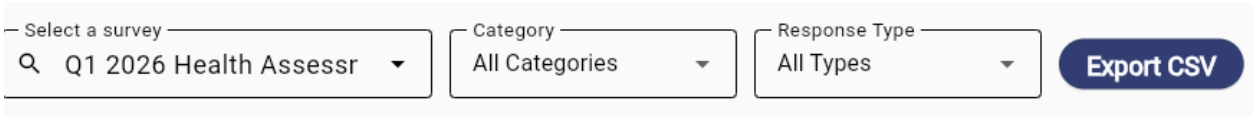
Data

Upon selecting this option at the top bar, you will be prompted with a rather empty page. The following can be seen below the bar, alongside a small text at the center of your screen prompting you to select a survey:



By Survey

This option allows you to select a singular survey to view the information of. In this case, the “Q1 2026 Health Assessment” survey which is filled with seeded data. This page can now be broken into three sections, the top section that holds your filtering information used to search for specific questions or to export the data as a CSV file:



A small section regarding the amount of respondents, how many people who were assigned to the survey that completed it, and the amount of questions. And the two viewing options, of “Data Table” and “Analysis”, which will display different information.

Data Table

The data table option will show the information within a table.

Data Table

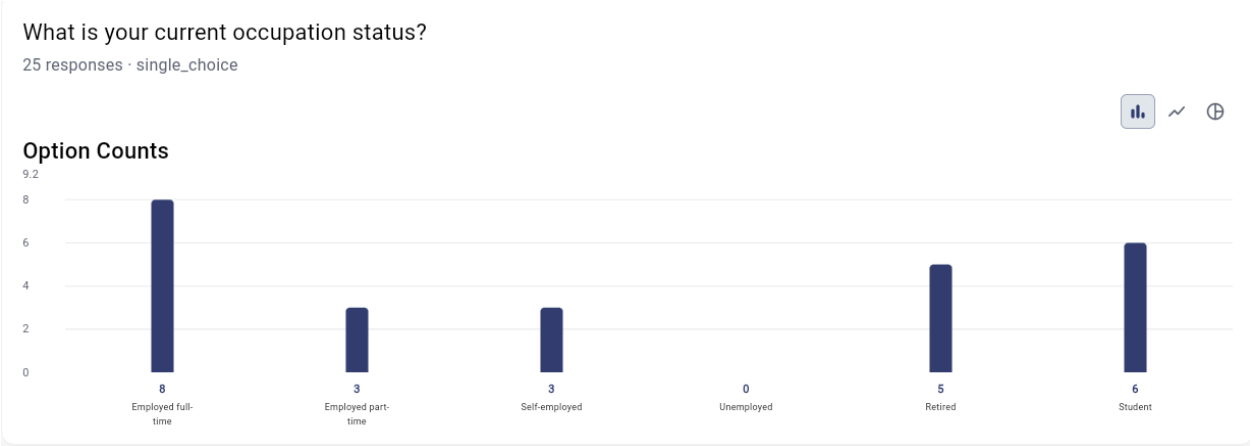
ID	What is your age range?	How do you identify your gender?	What is your current occupation status?	How many days per week do you exercise for at least...	Do you have any chronic health conditions?
X-741f8667	35-44	Female	Employed full-time	4	None
X-42362ec1	25-34	Male	Student	4	None
X-db14bfc9	35-44	Male	Employed full-time	3	None
X-8cf51cb7	25-34	Female	Employed full-time	5	None
X-568d13c6	25-34	Female	Student	5	None
X-ac845e24	45-54	Male	Employed full-time	2	Hypertension
X-1de0e4d8	55-64	Male	Retired	4	Hypertension

There will always be an ‘ID’ column at the left side, which relates to a singular participant without showing their name as to obfuscate the identity of the respondent. Each column thereafter will be a singular question, and the cells below will contain the answer that said participant gave.

Utilizing the filtering options above, the category and response type, you can filter the information shown. Categories will be broad things like demographics, mental health, physical health, and other categories tied to the questions. Response type will be specific to the question response type, whether it’s a single choice, multiple choice, a scale, a singular number, etc.

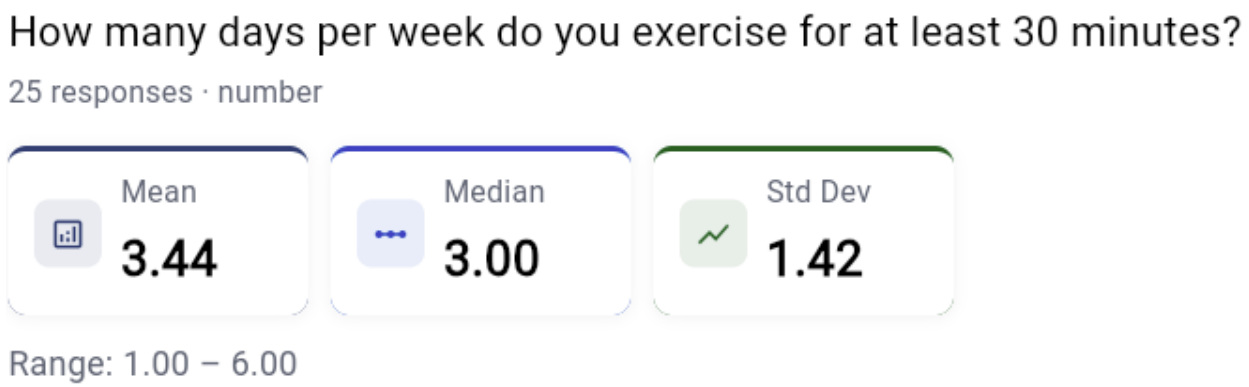
Analysis

Next we have the analysis page, which allows for a more visual experience with seeing data at a glance. Each question is now broken into a small section of the page, and shows a graphical representation of the data gathered from each question. For the following example, we'll be using the "Current Occupation Status" question of the same survey.



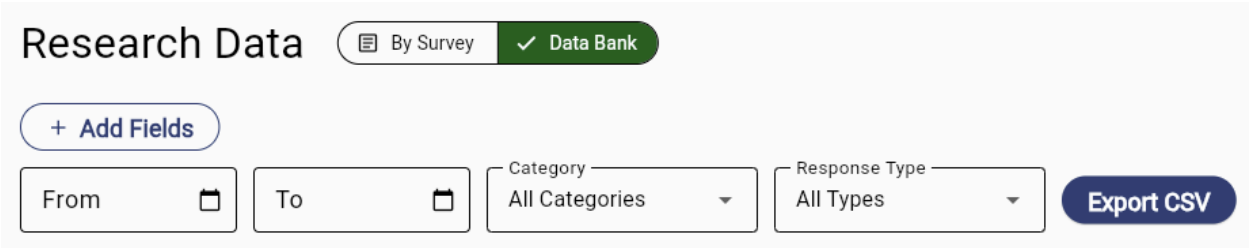
As a default, each question information is displayed as a bar chart. But, this can be changed on the right-hand side, with current options for a line chart and pie chart available shown below for the same question:

For other questions, if the type allows for it, the mean, median, and standard deviation of the responses will be given:



By Data Bank

The Data Bank option in the Data page is comparatively more complex to the ‘By Survey’ option. Where the ‘By Survey’ option only covers a single survey’s information, the Data Bank option instead filters by date. Giving you the following filter features:



The ‘Add Fields’ button will bring up a page that allows for you to select different questions from Surveys to add to the Data page. These selections will then appear on the data analysis section, with the same options for the Data Table and the Analysis page.

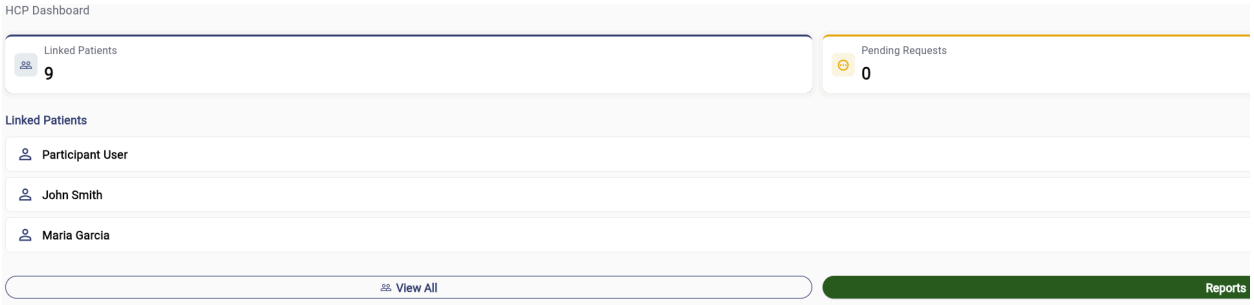
All selected fields will be displayed just under the filtering categories, and can be removed at any time or you may select ‘clear’ to completely clear your data. The information shown will also include the number of surveys you have selected, as there can be multiple instances of surveys if the researchers wish for a repeating one. The total respondents, the total number of questions, and the average completion rate of the surveys.

Otherwise, the Data Table and Analysis pages are the same as the ‘By Survey’ section. The real reason to use the Data Bank section is to amalgamate multiple survey answers into a single dataset. Any data selected, when selecting ‘Export CSV’, will be contained as shown. So the Data Bank setting allows for you to create specific datasets to your liking.

Healthcare Provider

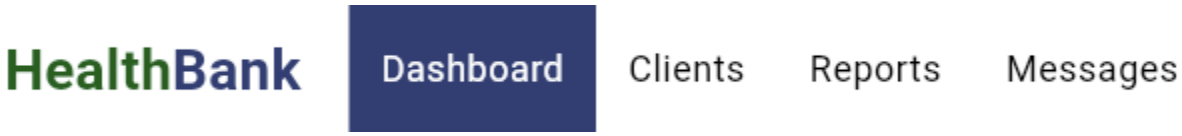
Dashboard:

Upon logging in, you will be routed to your Healthcare Provider dashboard. This dashboard will have several pieces of information on it, shown below.



It will have a number of ‘linked’ persons, which are the users whose information you will be able to see. The number of pending requests that you have submitted for linking a person (which is done through the ‘reports’ button found on the bottom right), and a list of linked participants.

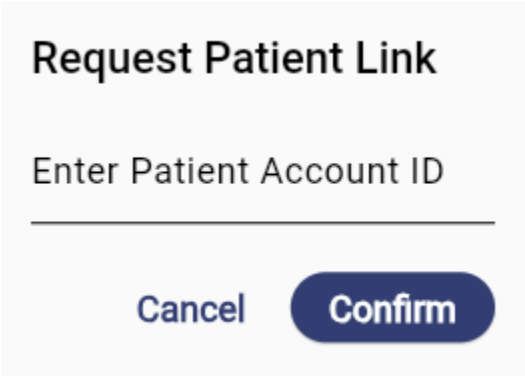
At the top, you’ll see four options:



Dashboard is what you’re currently on, and is the default on logging in.

Clients:

The clients page lists out your currently linked clients, and any pending requests you may have currently. With an option on the right-hand side to request patient link. Clicking on the button will give you the following popup:

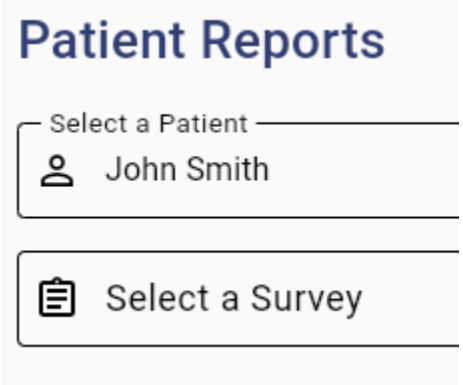


You enter the patient's account ID and hit confirm. The request is then sent off to an administrator for final confirmation.

Reports:

Next is the reports page, where you can view your linked clients' answers to different surveys to gather information that you would like to know.

You will select a patient by name, then will be prompted again to select a survey.



Upon hitting the ‘select a survey’ button, a list of all surveys your client has filled out will appear and you can select any of them. Upon selection, the questions asked on the survey and the answer given will be listed out below:

Patient Reports

Select a Patient

John Smith

Select a Survey

Q1 2026 Health Assessment

Q1 2026 Health Assessment

What is your age range?

45-54

How do you identify your gender?

Male

What is your current occupation status?

Employed full-time

How many days per week do you exercise for at least 30 minutes?

2

Do you have any chronic health conditions?

Hypertension

Please rate your current pain level

4

When was your last general health checkup?

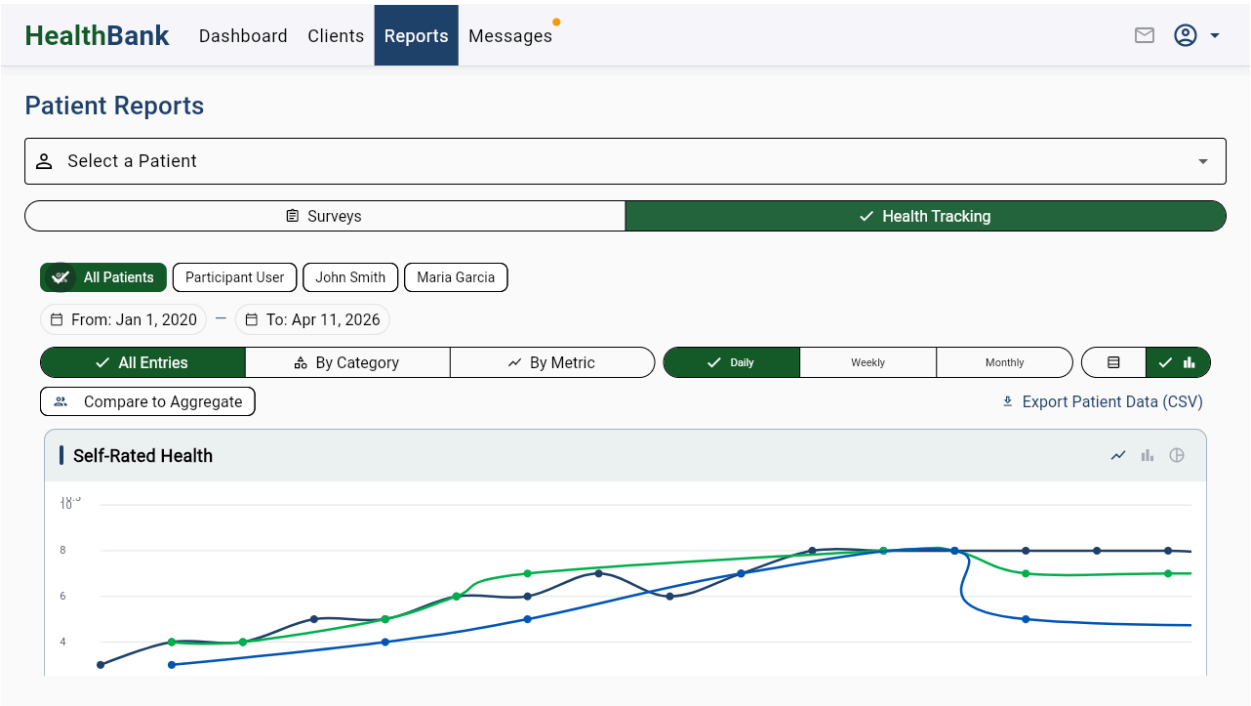
6-12 months ago

Health Tracking:

This section describes the Health Tracking tab added to the Reports page. It supplements the existing Healthcare Provider "Reports:" subsection of the manual.

Health Tracking Tab

The Reports page now contains two tabs: "Surveys" (existing functionality) and "Health Tracking" (new). Switching to the Health Tracking tab allows the healthcare provider to view tracking data for an individual patient.

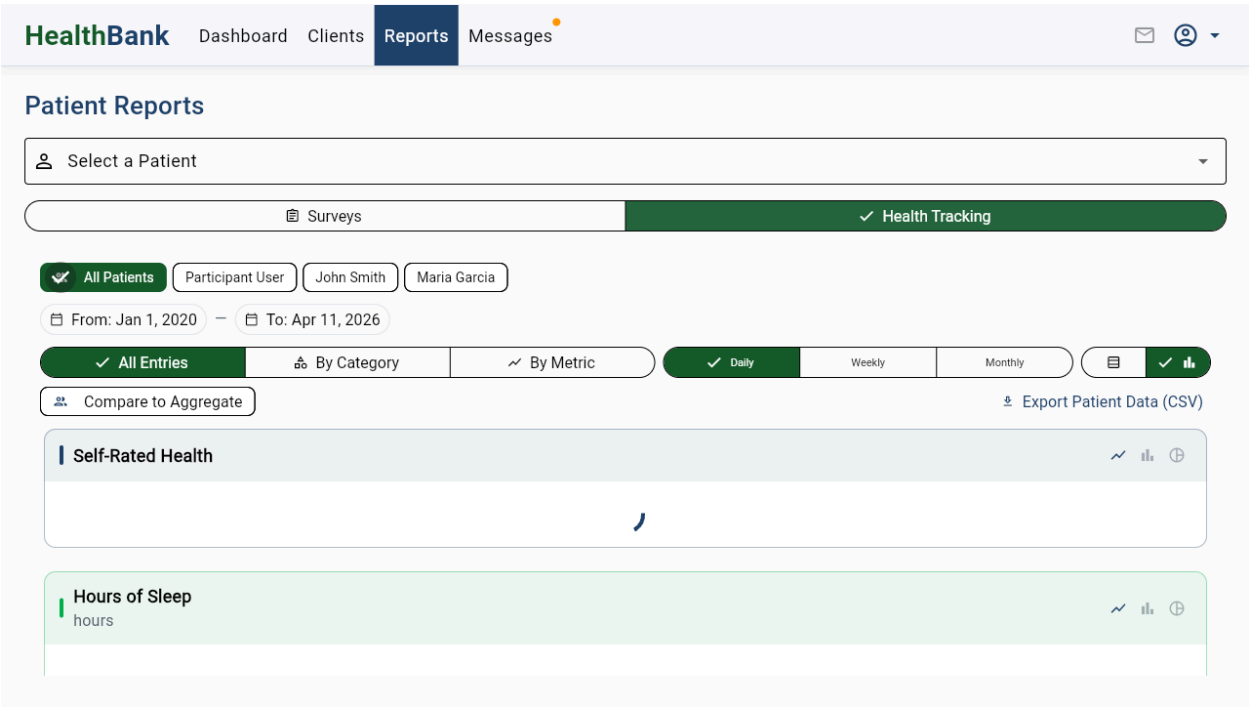


HCP Reports — Health Tracking tab with patient selector

To view data, first select a patient from the "Select Patient" dropdown at the top of the page. This dropdown lists all patients who have an active link with the provider and have consented to sharing their data. If a patient does not appear in the list, they may not yet have an active provider link or may not have granted consent.

Once a patient is selected, a metric filter panel appears. Select the desired metrics and date range, then click "Load Charts" to fetch the data. Each chart card

shows the selected patient's actual recorded values over time. A data table below each chart shows the exact values and dates.



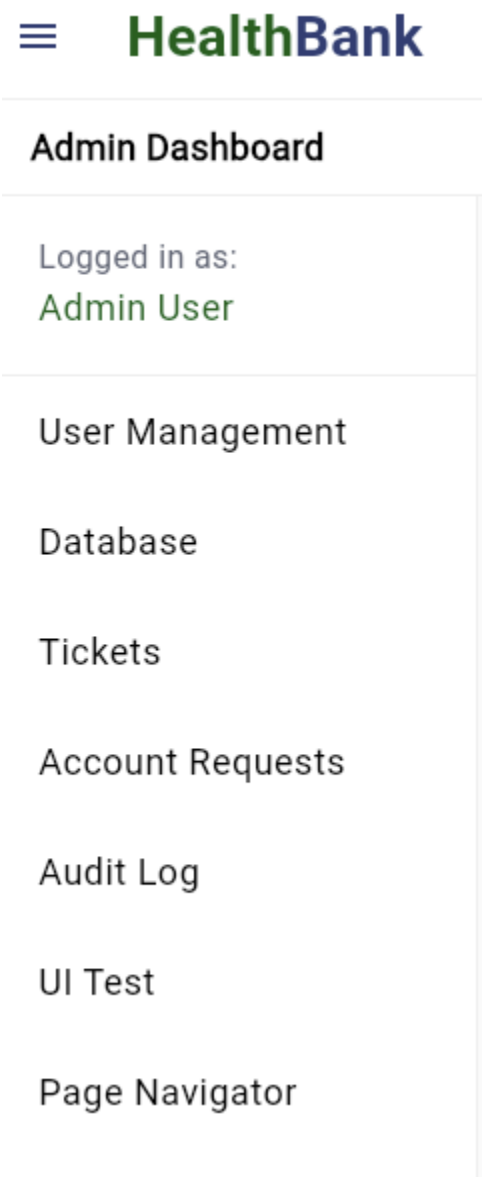
HCP Reports — patient health tracking chart

A "Comparison" mode toggle is available at the top of the chart area. When enabled, a population average overlay is added to each chart, allowing the provider to compare an individual patient's trajectory against the broader anonymized population trend. An "Export CSV" button downloads the current patient data as a CSV file.

Administrator Overview

In this section, it will show the abilities for the Administrator of the HealthBank application. It is recommended that any administrators also read over the Researcher and Healthcare Provider sections.

Upon first logging in, you will be put to the administrator dashboard with the following sidebar:



Unlike other users, all information for the Administrator is found on the dashboard. Selecting any of these options will change the view found on the majority of the page, which will be your working environment.

User Management:

Upon selecting ‘User Management’ the user manager will appear. At the top of the dashboard will be a search function, a filtering function that can be used to filter by roles of all users, and an active toggle button to filter between active and inactive users. Inactive users have had their accounts disabled by an administrator.

Below will be the following table (All names are seeded data, and will be removed upon final implementation):

Name	Email	Role	Status	Last Login	Actions
A Adam van Omme	avanomme@gmail.com	Researcher	Active	Never	
A Adam VO	adamvo@gmail.com	Researcher	Active	6d ago	
A Adam Vo	avanomme@upei.ca	Admin	Active	6d ago	
A Admin User	admin@hb.com	Admin	Active	1h ago	
C Charles Harris	charles.harris@email.com	Participant	Active	Never	
C Christopher Walker	christopher.walker@email.com	Participant	Active	Never	

This is a list of all current users of the HealthBank application. It lists their name, their email address that is linked to their account, their role (Admin, Researcher, Participant, and Healthcare Provider), their status of active or inactive, their last login, and administrator actions.

The administrator actions from left to right allow you to ‘view as user’, meaning you spoof their account and are able to view the website as if you were said user. Reset the password of the person. Edit the user's information (such as name, birthday, and role). Deactivate the user, meaning they won’t be able to do anything with their account. And finally delete a user, if a user account is no longer needed.

At the bottom, just below the table, there will be two numbers. A number of users shown, and the total number of users in the database.

Database:

The database tab allows you to view the database schema and held data without needing to pry into the backend. The following information will be shown at the top:

This shows you the number of databases within the table (as of time of writing, there are 27). Allows you to select a table, and either allows you to see either the schema or the data of the selected table. The ConsentRecord table will be used as an example:

ConsentRecord
Records of participants signing the consent document
10 columns

Column	Type	Constraints	Reference
ConsentRecordID	int	PK NN	-
AccountID	int	FK NN	AccountData(AccountID)
RoleID	int	FK NN	Roles(RoleID)
ConsentVersion	varchar(32)	NN	-
DocumentLanguage	varchar(5)	NN	-
DocumentText	longtext	NN	-
SignatureName	varchar(128)	NN	-
SignedAt	datetime	NN	-
IpAddress	varbinary(16)		-
UserAgent	varchar(512)		-

Primary Key Foreign Key Nullable Not Null

All columns, their data type, their constraints, and their reference key is shown. For constraints, a legend is provided at the bottom.

By viewing as a table, a table format will be shown:

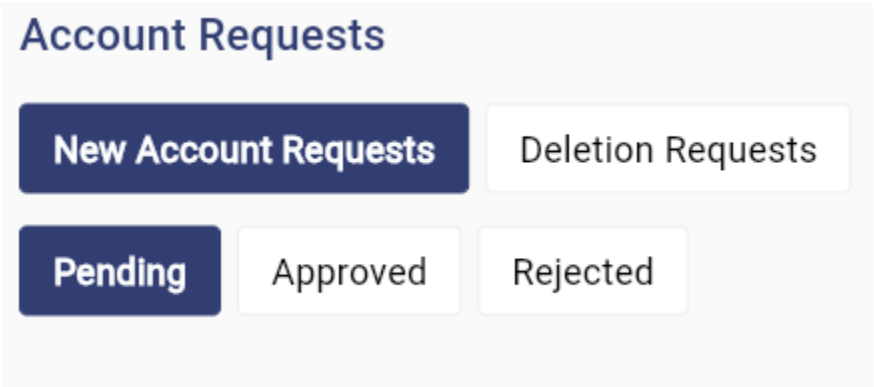
ConsentRecord Data 33 rows total

ConsentRecordID	AccountID	RoleID	ConsentVersion	DocumentLanguage	Doc
1	1	4	1.0	en	I agr
2	2	1	1.0	en	I agr
3	3	3	1.0	en	I agr
4	4	2	1.0	en	I agr
5	5	2	1.0	en	I agr
6	6	2	1.0	en	I agr
7	7	2	1.0	en	I agr
8	8	2	1.0	en	I agr
9	9	2	1.0	en	I agr
10	10	3	1.0	en	I agr
11	11	3	1.0	en	I agr

This shows the information within the table as it is stored, by column. Though the data cannot be exported from this page.

Account Requests:

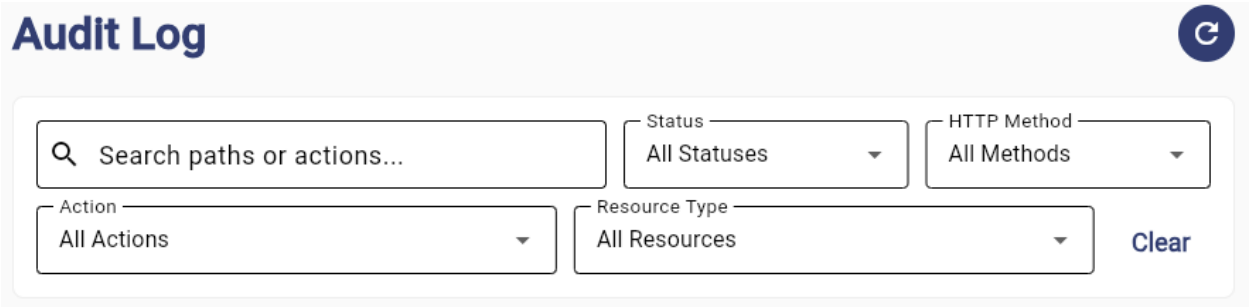
The account requests page is where administrators approve the creation of new accounts. With the following buttons:



Each button will show what it says. Any new account requests that are pending to be approved, a list of approved new account requests, and a list of rejected new account requests. There is also the same for account deletion requests.

Audit Log:

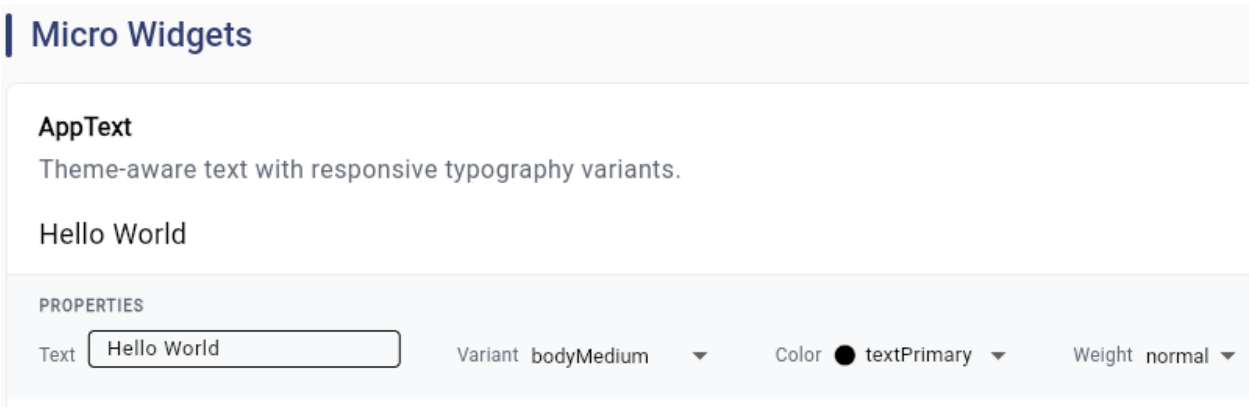
The audit log keeps a track of all requests to the database by all types of users. With the following search options available:



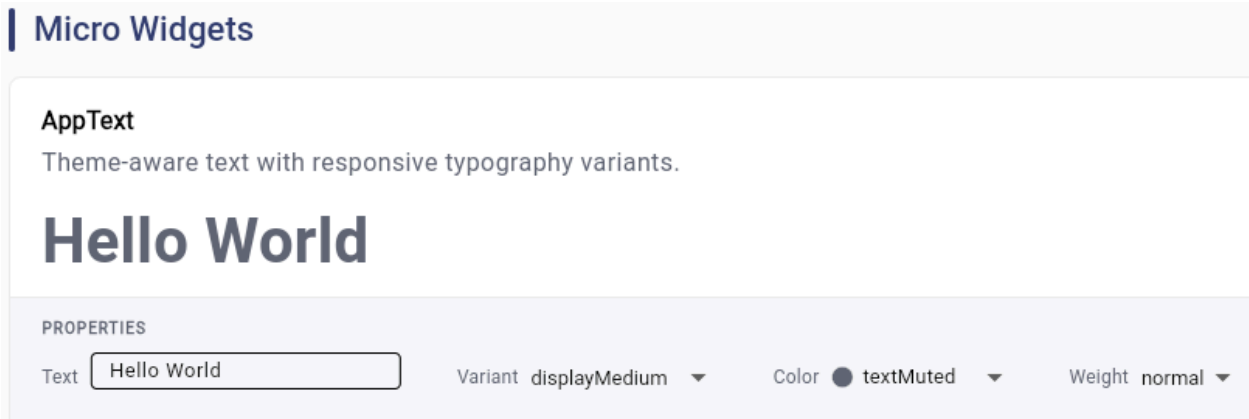
These options allow for a narrowing of information of requests, to search for any oddities or potential bugs that an administrator may need to fix. All information is laid out in a table format below.

UI Test:

The UI test page is where administrators can test small parts of frontend widgets without needing to try implementation first. Each widget will have a pre-set amount of options available. This example will use text:



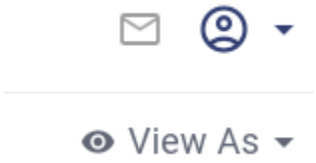
For app text, you can change the words displayed, the size of the text, the colour, and the text weight in real time:



There are many options here in this section, and it doesn’t effect anything any user sees. You can see modifications for things such as buttons, text, forms, or information in popups. This is mainly to be used if any small changes are needed to formatting or for visual clarity.

View As:

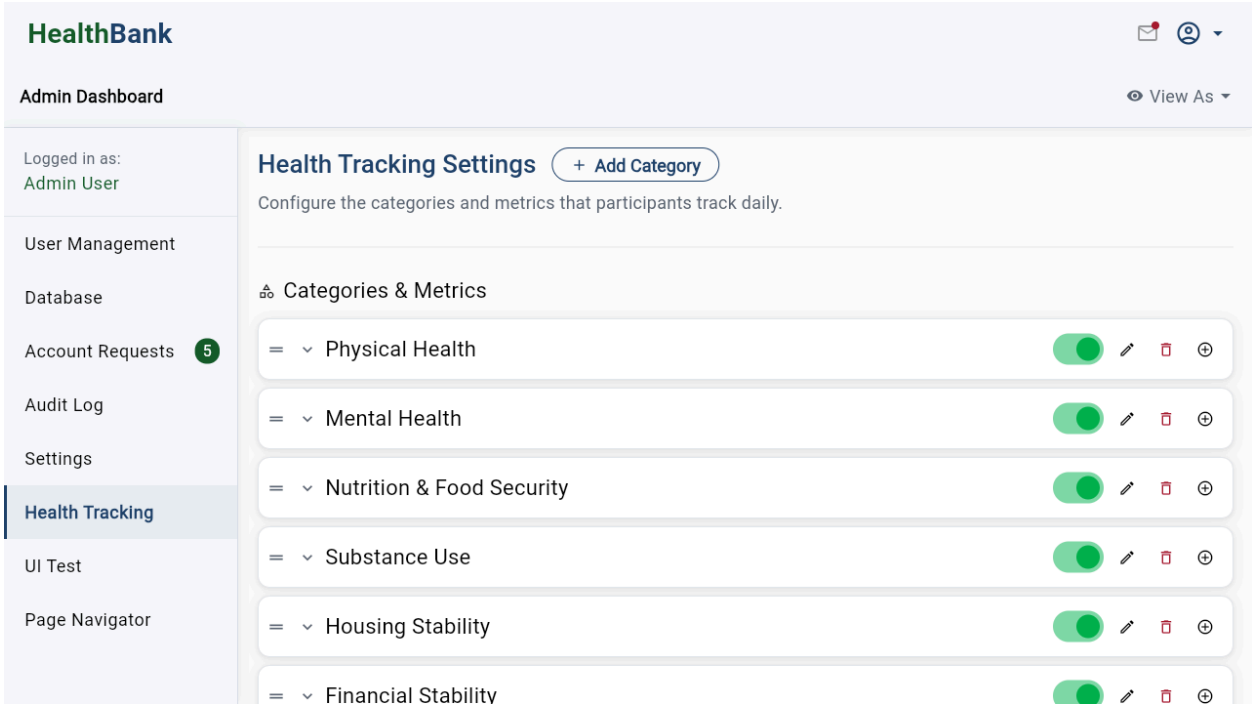
The final functionality of the administrator is the ‘view as’ button found in the top right of your dashboard, just below your profile:



This button will prompt you with either Participant, Researcher, or Healthcare Professional. Upon selecting any option, you will be taken to a page of the respective role with all their functionality and limitations. This allows administrators to act as different roles temporarily, which can be extremely useful in bug fixes or implementation of new functionality.

Health Tracking Settings:

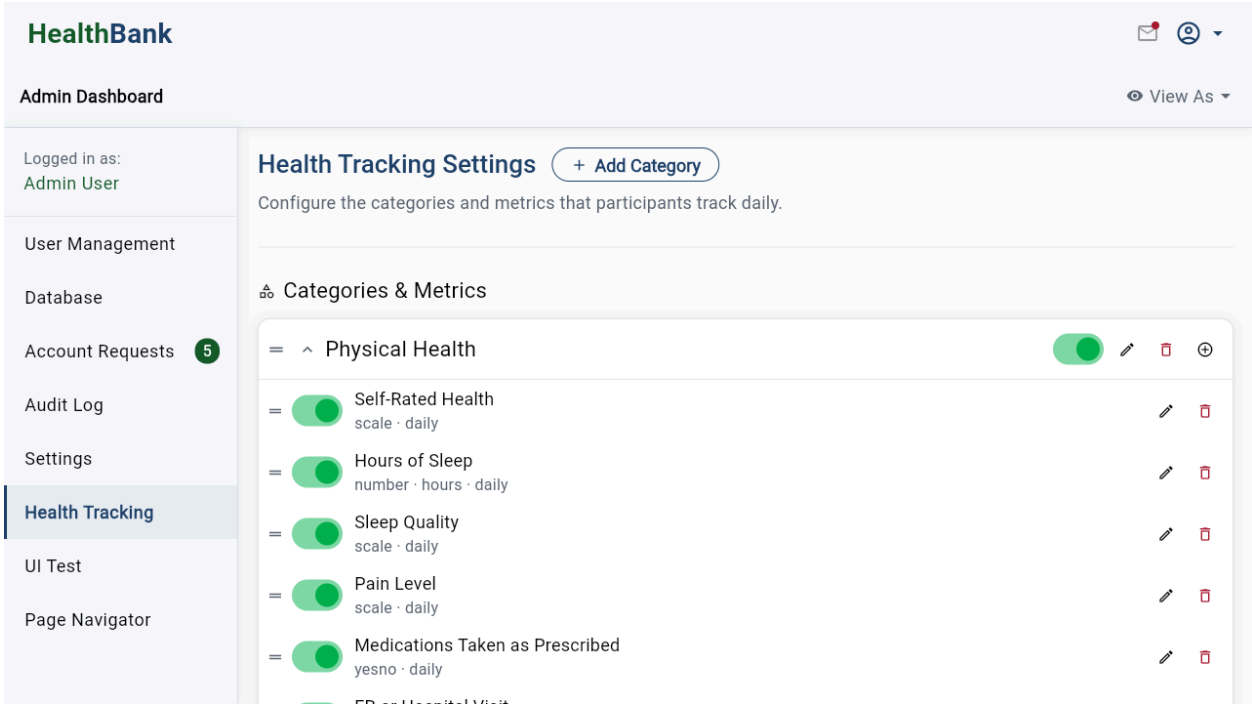
The Health Tracking Settings page is where administrators configure the categories and metrics that appear on the participant Health Tracking page. Navigate to it by selecting "Health Tracking" from the admin sidebar.



Admin Health Tracking Settings — category list with active toggles

The page displays a list of all tracking categories. Each category row shows the category name and can be expanded to reveal the metrics within it by clicking the row. An "Add Category" button in the top-right corner opens a dialog to create a new category. The following actions are available for each category and metric via the icons on the right of each row:

- Active toggle — enables or disables the item. Inactive metrics no longer appear on the participant Log Today page.
- Edit (pencil icon) — opens a dialog to change the name, description, metric type, unit, scale range, choice options, and collection frequency.
- Reorder — categories and metrics can be dragged and dropped to change their display order.
- Remove (trash icon) — soft-deletes the item. Removed items move to a "Removed Items" section at the bottom. Historical data is preserved.

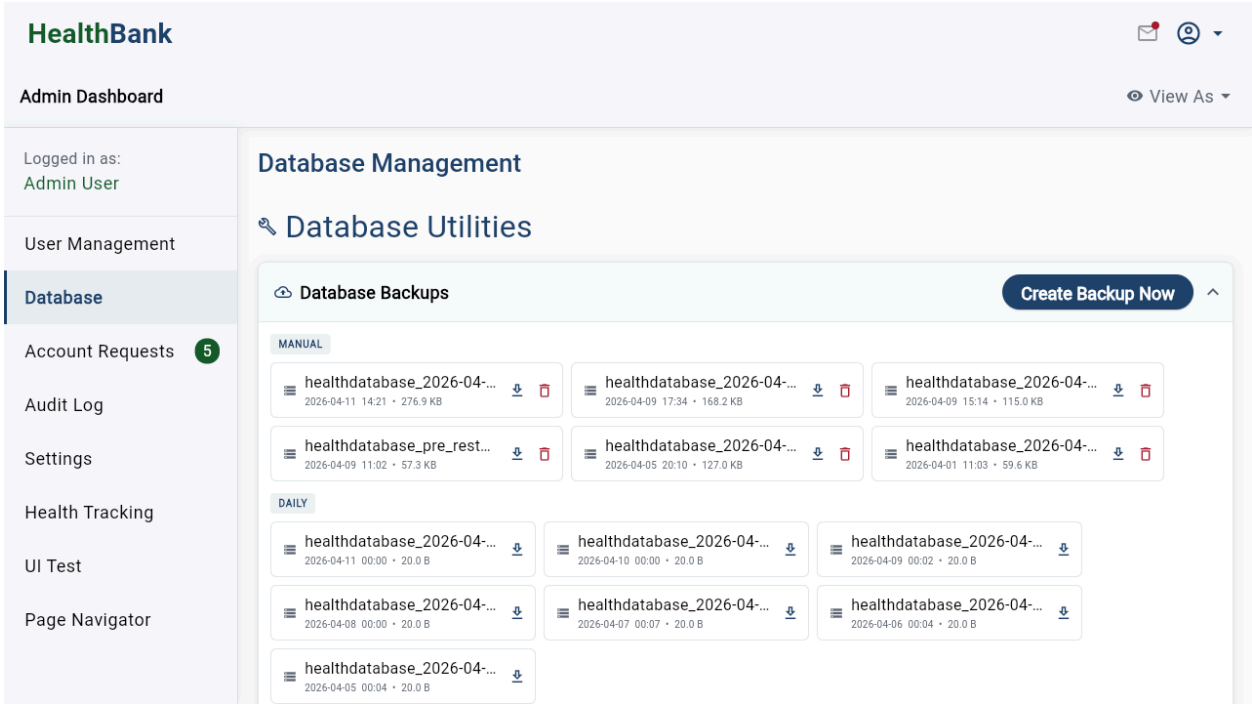


Admin Health Tracking — Physical Health category expanded showing its metrics

When adding or editing a metric, the available fields are: display name, description, metric type (scale, number, yes/no, single choice, or text), unit (for number types), scale minimum and maximum (for scale types), choice options (for single-choice types), and collection frequency (daily, weekly, or monthly). All changes take effect immediately after saving.

Database: Backup & Restore

The Database page now includes a "Database Utilities" panel above the existing table viewer. This panel provides manual backup creation, download, restore, and deletion. This supplements the existing "Database:" subsection of the manual.



Admin Database — Database Utilities backup panel showing existing backups

A "Create Backup Now" button at the top-right of the panel triggers an immediate database backup. Once complete, a success notification shows the size of the backup file and the list refreshes automatically.

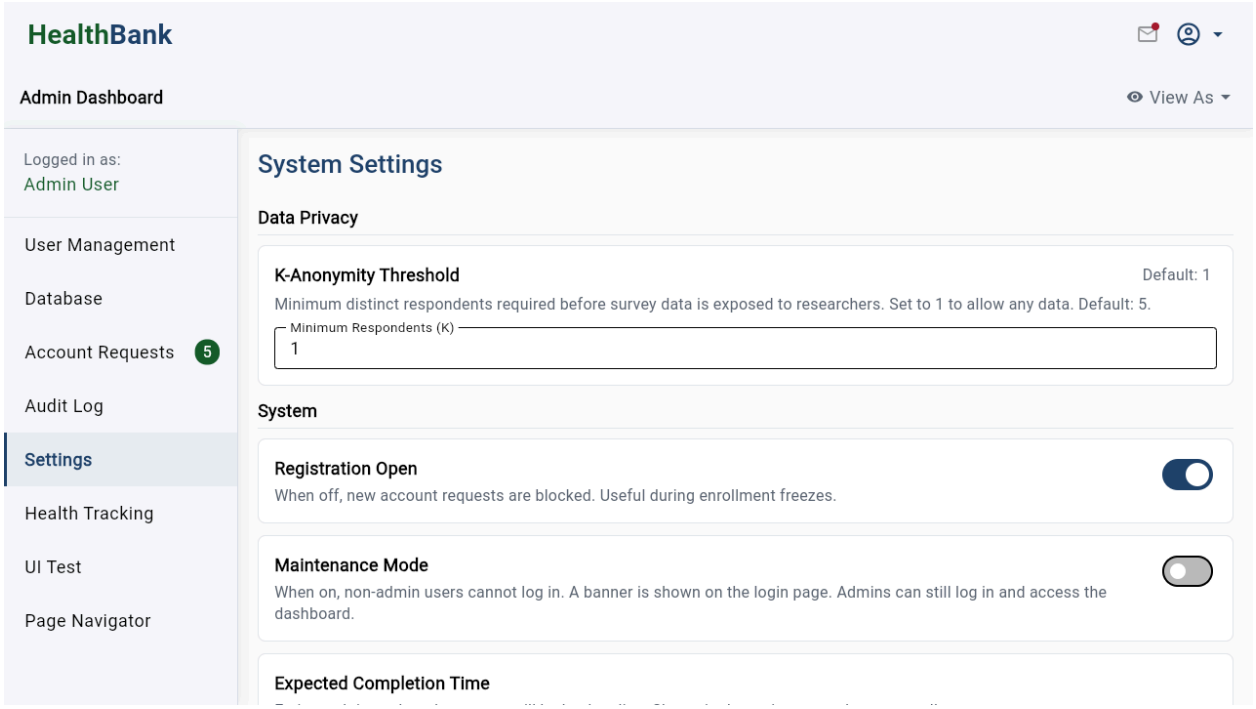
The backup list shows all available backups with the filename, creation date and time, file size, and the following action buttons:

- Download — downloads the backup file. On the web application the browser download is triggered; on mobile the file is saved to device storage.
- Restore — restores the database to the state captured in the selected backup. A confirmation dialog is shown first. The process automatically creates a pre-restore safety backup, applies the backup file, and runs any pending migrations. A success notification reports the pre-restore size and migrations applied.
- Delete — permanently removes the backup file from the server after confirmation.

Caution: restoring from a backup will overwrite all current data in the database. The automatic pre-restore backup provides a safety net, but this action should only be performed when necessary.

Settings:

The Settings page provides administrators with control over system-wide operational parameters. Changes take effect across all users almost immediately. Navigate to it by selecting "Settings" from the admin sidebar.



Admin Settings page — Data Privacy, System, and Login Security sections

The settings are organised into three sections:

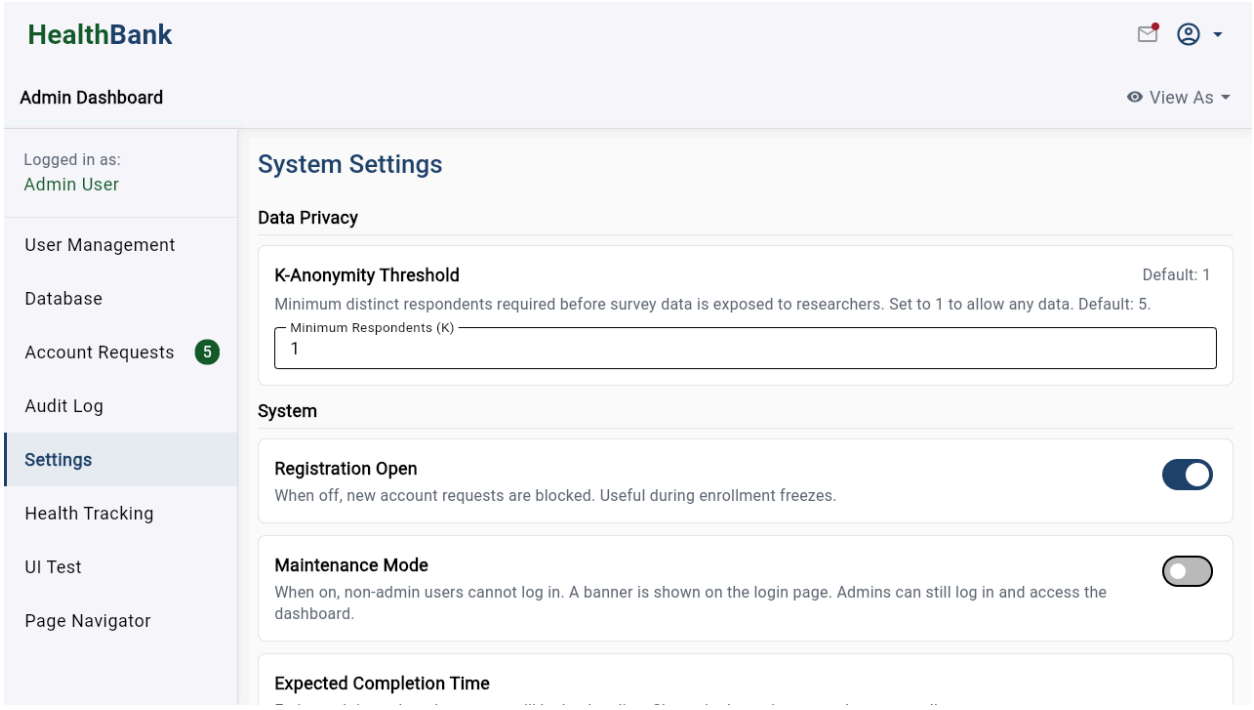
Data Privacy — contains the K-Anonymity Threshold field. This sets the minimum number of participants that must contribute to a data point before it is shown in researcher or HCP analytics. Points with fewer contributions are suppressed. A higher threshold provides stronger privacy protection.

System — contains three toggles and a text field:

- Registration Open — when enabled, new users can submit account requests via the login page. When disabled, the "Request an Account" button is hidden.
- Maintenance Mode — when enabled, a maintenance banner is displayed at the top of every page to notify users of scheduled downtime.
- Maintenance Completion — sets the expected completion time shown in the maintenance banner (e.g. "3:00 PM today"). Only visible to users when Maintenance Mode is active.
- Consent Required — when enabled, participants must sign the consent document before accessing survey tasks or health tracking.

Login Security — contains two numeric fields:

- Maximum Login Attempts — failed attempts allowed before an account is temporarily locked. Set to 0 to disable lockout entirely.
- Lockout Duration — minutes an account remains locked after exceeding the maximum failed attempts.



Admin Settings — Save Settings button

Boolean toggles save automatically when changed. Numeric and text fields require clicking the "Save Settings" button at the bottom of the page. A toast notification confirms success or failure.